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Risk aggregation, evolution, and contagion of different dimensions among industries in China: based on coupled network analysis --Manuscript Draft--

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Abstract:	Does inter-industry risk contagion in China operate through a single channel? We argue it does not. Traditional models often mix supply-chain transmission with financial leverage amplification. Using a five-dimensional coupled network framework across 31 Chinese industries from 2007 to 2023, we document a dual-track mechanism supported by the near-zero correlation between price and tail risks. Our results show that the dominant contagion channel switches depending on the market regime: (1) The entity-driven channel (measured by price risk) functions as a structural backbone. It reflects continuous supply-chain transmission and maintains stable dominance during both normal periods and real-economy shocks (e.g., trade friction), with technology sectors acting as risk absorbers; (2) The financial-driven channel (measured by tail risk) is crisis activated. Typically dormant, it erupts explosively during leverage-fueled crises (e.g., the 2015 crash), characterized by bidirectional amplification loops and the shift of non-bank financial institutions to core spillover roles. Mechanistic analysis reveals that this financial fragility is rooted in a structural “finance-property closed loop”, where economic financialization and real estate dependence disproportionately amplify tail risk contagion. Consequently, effective regulation demands a dual-track approach: routine surveillance for supply-chain risks and preemptive measures for leverage-driven financial contagion.

- This study explores the dual-track risk contagion mechanisms across industries.
- Crisis type determines which channel activates.
- Entity-driven channel maintains stable as a structural backbone via supply-chain.
- Financial-driven channel remains dormant but erupts during leverage-fueled crises.
- This financial fragility is rooted in a structural “finance-property closed loop”.

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1. Introduction

Does inter-industry risk contagion in China flow through a single channel? We argue it does not. Consider two typical shocks: a surge in production costs for a steel manufacturer (a real-economy shock) versus a leverage collapse in a securities firm (a financial-sector shock). The former transmits primarily through supply chains, whereas the latter spreads via financial leverage. However, traditional risk contagion models, often based on single-dimensional frameworks like volatility spillovers (Diebold and Yilmaz, 2014), tend to mix these distinct mechanisms. This overlooks a fundamental empirical reality: distinct risk dimensions—specifically price changes versus tail events—can be statistically orthogonal and operate through different network structures. Treating these dimensions as interchangeable fails to capture the heterogeneous nature of systemic risk, posing challenges for risk management in emerging markets where the divergence between real-economy and financial-sector shocks is particularly pronounced.

As the world’s second-largest economy and a representative emerging capital market, China exhibits unique institutional features that require a tailored analytical framework. First, China’s ongoing transition from a bank-dominated system to a multi-tiered capital market has diversified

cross-sector risk contagion channels while often obscuring their specific pathways (Carpenter et al., 2021). Second, the policy-driven economic paradigm links industrial policies directly with market behaviors, creating complex risk transmission patterns across sectors (Zhou et al., 2025). Third, structural constraints, such as the dominance of retail investor (over 60% market participation), uneven information disclosure quality, and underdeveloped derivatives markets, amplify the asymmetry of risk propagation. These characteristics make China an ideal laboratory to investigate how different risk dimensions operate through distinct channels, offering insights with both regional and global implications given China's systemic spillover potential in Asia's financial connectivity (Glick and Hutchison, 2013; Morelli and Vioto, 2020).

Our central thesis is that inter-industry risk contagion in China operates through two fundamentally distinct tracks. The first is a structural, entity-driven channel, measured by price risk, which continuously transmits shocks via supply chains. The second is a financial-driven channel, measured by tail risk. This channel typically remains dormant during normal periods but becomes the dominant force during systemic crises. We argue that these two channels differ significantly in their timing, network structure, and economic sensitivity. Specifically, the entity-driven channel reflects the steady transmission of costs and demand along production networks. In contrast, the financial-driven channel is triggered by extreme events, such as the 2015 stock market crash, driven by leverage unwinding, liquidity freezes, and market panic.

To empirically disentangle these dual tracks, we address three limitations in existing literature. First, prior studies mainly focus on risk contagion within financial sectors (Nguyen et al., 2020) or across national stock markets (Abuzayed et al., 2021). This leaves a gap in understanding the spillover mechanisms between the financial sector and the real economy. Given the close link between financial market volatility and real economic performance, coupled with risk transmission through trade networks (Acemoglu et al., 2012), the analysis must extend beyond financial subsectors. To address this, we constructed a comprehensive dataset covering 31 sectors in China, spanning both financial industries (such as banking and non-bank finance) and real-economy sectors (such as manufacturing and technology), to systematically investigate these cross-sector risk transmission chains.

Second, few studies combine dynamic and static perspectives to characterize how risk contagion evolves over time, especially during market transitions. Capturing this temporal variation requires a framework that can identify both persistent structural patterns and sudden, crisis-specific dynamics. Therefore, we implement a dual strategy: (1) a dynamic analysis using rolling-window networks to track the continuous evolution of risk contagion from 2007 to 2023, revealing how the dominance of risk dimensions shifts; and (2) a static analysis focusing on five key historical periods—such as the global financial crisis and the China-US trade friction—to isolate specific contagion patterns through case studies.

Third, existing network approaches often rely on linear frameworks and traditional Granger causality tests. These methods struggle to capture the nonlinear characteristics, structural breaks, and asymmetric dependence patterns inherent in financial systems (Shahzad et al., 2018; Chen and Sun, 2024). Moreover, traditional unidimensional frameworks based on volatility or correlation metrics (Diebold and Yilmaz, 2014) fail to capture how transmission mechanisms diverge across different risk dimensions during extreme events.

To overcome these limitations, we develop a five-dimensional risk framework integrating price risk, volatility risk, asymmetric risk, extreme risk, and tail risk. Our approach involves three key steps: (1) using multivariate nonlinear Granger causality tests with a leave-one-out (LOO) approach to construct distinct subnetworks for each risk dimension (Hué et al., 2019); (2) applying network coupling methods to estimate time-varying weights that quantify the dominance of each dimension in driving systemic risk (Bonaccolto et al., 2019; Billio et al., 2023); and (3) calculating edge uniqueness to identify contagion paths specific to a single dimension (Musmeci et al., 2017). This framework enables us to test our central hypothesis: if the two channels are distinct, the price risk network (entity channel) should show persistently high uniqueness, while the tail risk network (financial channel) should exhibit explosive surges in uniqueness and influence only during crises.

Our empirical findings support the dual-track hypothesis through three layers of evidence. First, dynamic network analysis establishes the temporal separation between the two channels. By decomposing the network, we find a fundamental distinction: Price risk networks (the entity channel) exhibit persistently high uniqueness and stable coupling weights across all periods. This confirms that supply-chain transmission operates continuously as a structural backbone. In contrast, tail risk networks (the financial channel) are crisis-specific. Their uniqueness and coupling weights are low during tranquil periods but surge explosively during the 2008 financial crisis and the 2015 stock market crash. Sectoral data supports this: cyclical industries maintain a stable risk contribution (25-35%), while the financial sector's contribution spikes (>40%) only during crises. Notably, during the COVID-19 pandemic, while tail risk uniqueness rose, its dominance (weight) remained moderate, confirming the pandemic as primarily a supply-chain shock rather than a leverage crisis.

Second, case studies validate that shock origin determines which channel activates. We analyze two contrasting periods. The China-US trade friction (2018-2020) represents a typical entity-driven shock. Risk propagated one-way through supply chains, where technology and export-oriented sectors acted as a "risk sink," accumulating stress transmitted from upstream industries due to direct trade restrictions, while tail risk remained subdued. Network topology during this period reflects precise, industry-specific transmission along production linkages—distinctly different from systemic financial contagion patterns. Conversely, the 2015 stock market crash represents a financial-driven shock fueled by excessive leverage. Here, tail risk became overwhelmingly dominant. Non-bank financial institutions shifted from peripheral roles to become core risk spreaders, and the roles of the real sectors are fragmented. This comparative analysis demonstrates that transmission channels are not static network properties but dynamically activated depending on whether shocks originate from real-economy disruptions or financial-system instabilities.

Third, mechanistic analysis identifies why the financial channel is so fragile. Having established that entity-driven and financial-driven channels exhibit distinct patterns, we investigate what macroeconomic factors render the financial channel explosively fragile during crises. Controlling for network effects, we find that economic financialization and real estate prosperity significantly amplify contagion intensity. Our analysis suggests these factors specifically fuel the financial-driven channel rather than the entity-driven one. This is evidenced by the asymmetry between periods: during the 2015 crash, high financialization levels coincided with the explosive activation of tail risk and the dominance of non-bank financial institutions. Conversely, during the trade friction period, these macro-financial vulnerabilities remained dormant, and risk followed

supply chains. This vulnerability stems from a structural "finance-property closed loop". High financialization concentrates leverage in shadow banking, while real estate credit links property cycles to systemic stability. Once triggered by panic or deleveraging, this loop acts as an amplifier, igniting the explosive contagion observed in our tail risk networks. In contrast, the entity-driven channel operates through physical production linkages. It lacks this leverage-based amplification mechanism, which explains why it maintains structural stability even when financial risks erupt.

These findings have profound implications for financial regulation and systemic risk prevention in emerging markets. Our evidence suggests that effective risk management demands a dual-track regulatory approach tailored to the distinct mechanisms of each channel: (1) For the entity-driven channel, policymakers should establish supply-chain-based early-warning systems, leveraging industry price indices to monitor structural and persistent risk transmission—a task of routine surveillance; (2) For the financial-driven channel, regulators must prioritize macroprudential policies targeting financial leverage and real estate financing to preempt the activation of tail risk contagion—a critical focus for crisis prevention. Ultimately, single-track regulatory frameworks that treat all risk dimensions equivalently will prove inadequate in addressing the heterogeneous nature of systemic risk in complex, interconnected financial systems.

The remainder of this paper is organized as follows. Section 2 reviews the literature on financial risk contagion, focusing on risk carrier, contagion dimensions, and identification methods. Section 3 details the dataset and the multidimensional network methodology. Section 4 presents the empirical results, including dynamic network evolution, static case studies, and mechanistic analysis. Finally, Section 5 concludes the paper with policy implications.

2. Literature review

This paper is closely associated with the study of financial risk, particularly the application of complex networks to the analysis of financial risk contagion. By categorizing risk carriers, contagion dimensions, and identification methodologies, the existing literature has conducted a systematic investigation into financial risk contagion.

From the perspective of risk carriers, most literature focuses on the contagion of financial risks in transnational stock markets. Risk indicators are constructed by using the return data of broad-based stock indexes of various countries in different periods, and risk transmission models are designed to analyze the transnational contagion mechanism of systemic risks, and the relationship between financial risk contagion channels and economic states is studied. For example, the risk information transmission mode under special periods such as the international financial crisis and the COVID-19 pandemic (Abuzayed et al., 2021). Some studies note the complexity of the composition of global financial markets and believe that the cross-market and cross-asset transmission path should also be included as an important factor affecting the stability of a country's financial market (Huang and Liu, 2023). However, financial risks will not only be transmitted across regions and markets, but there is a more refined risk aggregation and contagion mechanism at the medium and micro levels of the industry (Yao and Jiang, 2025). Risk events in a single industry have strong externalities, and the chain reaction of risk contagion between industries will lead to the outbreak of systemic risks (Acemoglu et al., 2012). Especially in the financial market, investors'

irrational imitation behavior may trigger market panic, expand the impact intensity of heterogeneous shocks, and form cross-sector and cross-industry financial risk contagion (Avery and Zemsky, 1998).

However, a large number of existing studies mostly focus on risk contagion among sub-industries of the financial sector, such as the exploration of risk contagion among different banks, securities brokers, and insurance companies (Yang et al., 2019; Zhao et al., 2025). These studies believe that the risks in the financial industry were the root cause of global risks, but did not include the real economy industry or the financial industry in the research scope. Different sectors can exert influence through balance sheet channels, especially between the real and financial sectors. Specifically, when a single entity or industry is faced with a crisis, enterprises will sell assets in a concentrated manner to ensure liquidity and cause a sharp depreciation of related assets, thus leading to a contraction of balance sheets within the industry (Carvalho, 2015). The decline in the value of mortgage assets will further aggravate the vulnerability of the financial sector; thus, the transmission path of risks in the real economy to the financial sector is formed (Giannetti and Saidi, 2019). Especially in recent years, due to the intensified changes in the international situation and the frequent occurrence of public emergencies, the supply relationship of the industrial chain between industries is extremely vulnerable to impact, which leads to the rapid spread of extreme risks in individual industries to the financial market and the real economy in a short period. This necessitates adopting cross-industry risk contagion as the focal point, integrating real-economy sectors and financial industries into a unified framework to more comprehensively and precisely characterize risk transmission mechanisms in markets.

From the perspective of contagion dimensions, traditional financial risk measurement primarily focuses on the mean and volatility of asset returns, namely the first and second moments. By investigating correlations between the first and second moments across assets, researchers analyze contagion pathways of financial risks. For instance, Asgharian and Nossman (2011) employed a stochastic volatility model to study risk spillover mechanisms from the U.S. stock market to European stock markets through the second-moment dimension. Yang and Zhou (2017) examined the impact of quantitative easing on global stock market volatility spillover patterns from the second-moment perspective. Gao et al. (2021) investigated time-varying first- and second-moment connectedness networks between Chinese green bonds and equities, commodities, foreign exchange, and money markets under financial event-driven scenarios.

With the continuous development of China's financial markets and the gradual introduction of short-selling mechanisms and derivatives, scholars have begun to focus on the dimension of tail risk (Ando et al., 2022). Compared to mean and volatility levels, tail risk exhibits stronger contagion effects and broader transmission scopes. Once major adverse events occur, tail risk propagates through economic and financial industry linkage networks, threatening the stability of the entire financial-economic system. In recent years, frequent extreme risk events and persistent volatility in international financial markets have revealed limitations in existing risk dimensions for comprehensively characterizing complex intermarket return distribution linkages and asymmetries. For instance, regarding the second moment, variance simplistically equates losses and gains, failing to fully represent real and refined risk measurement, as downside risks typically exhibit higher realization probabilities than upside risks (Bouri and Jalkh, 2023). In light of this, emerging research suggests that a substantial portion of risk information transmission can be reflected through higher-

order moment spillovers. There is growing consensus on incorporating asymmetric risk and extreme risk dimensions, utilizing kurtosis and skewness characteristics of financial asset return distributions to capture extreme event frequencies and pricing asymmetries (Neuberger and Payne, 2021). Theoretically, kurtosis spillovers reflect the cross-market propagation of extreme event risks, while skewness spillovers measure the transmission of return distribution asymmetries across markets (Bouri et al., 2021; Bouri and Jalkh, 2023).

While existing studies have increasingly adopted complex network methodologies, real-world risk contagion across financial markets or real economic sectors often manifests more intricate patterns. These processes typically exhibit multi-layer network structures characterized by multi-dimensional, hybridized interconnections and time-varying complex linkage effects (Bargigli et al., 2015). Multiple forms of risk contagion may erupt simultaneously, with sophisticated interaction mechanisms operating across different network layers. Consequently, risk contagion networks constructed solely on single dimensions fail to reflect authentic transmission mechanisms or accurately capture real-world risk diffusion patterns. Instead, researchers should employ scientific approaches like coupled networks to develop multi-layer synthetic frameworks that capture cross-layer interaction dynamics and identify complex financial risk transmission pathways (Wang et al., 2021).

From the perspective of identification methodologies, on one hand, to verify the existence of intermarket risk contagion relationships, (Hong et al., 2009) pioneered the application of Granger causality tests in financial risk measurement, examining linear risk spillovers between pairwise financial assets. However, early-stage studies predominantly focused on pairwise interaction mechanisms between financial entities, neglecting the holistic network connectivity of financial risks, thereby failing to accurately characterize systemic shocks emerging within financial networks (van de Leur et al., 2017). When research scope expands to multiple financial assets or real-sector industries, this necessitates capturing both the heterogeneity and interconnectedness of risk transmission processes. Consequently, complex network models have garnered increasing scholarly attention, offering robust analytical tools for delineating risk correlations among financial entities and assessing financial system fragility (Denbee et al., 2021). These network-based approaches enable the identification of systemically important nodes, measurement of risk propagation channels, and detection of contagion path dependencies through topological analysis of financial network architectures.

However, existing literature employing network correlation approaches to investigate financial risks predominantly operates within linear frameworks combined with traditional Granger causality testing methods. In financial systems, asset return series often exhibit significant nonlinear characteristics and structural breaks due to factors such as fire sales of commonly held assets (Kyle and Xiong, 2001; Duarte and Eisenbach, 2021), panic-driven systemic risk contagion among investors (Barunik et al., 2020), and asymmetric dependence between asset returns during market upswings versus downturns (Bollerslev et al., 2022). Furthermore, risk spillover effects demonstrate pronounced asymmetry in their responses to changing market conditions, with financial risk contagion frequently manifesting nonlinear propagation patterns (Shahzad et al., 2018; Chen and Sun, 2024). To address these complexities, researchers have incorporated multivariate nonlinear Granger causality methods into risk contagion network construction, accounting for the tail

clustering effects and asymmetric nature of financial risks (Diks and Wolski, 2016). Additionally, when analyzing networks containing numerous nodes, scholars have introduced the leave-one-out approach to enhance causal relationship identification amidst substantial third-party interference (Zedda and Cannas, 2020). This methodology systematically calculates causal relationships while controlling for each third-party market, thereby improving network structure characterization.

On the other hand, the majority of existing studies utilize vector autoregressive (VAR) models to investigate influencing factors of market risk contagion after constructing risk transmission networks (Huang and Liu, 2023). Some research enhances the basic VAR framework with advanced estimation techniques, including error correction models, frequency domain decomposition methods, and time-varying coefficient approaches (Favero, 2013; Antonakakis et al., 2018; Baruník and Křehlík, 2018). However, traditional VAR models face inherent limitations in handling excessive node quantities and time-varying contagion structures within networks, often leading to unreliable estimation outcomes when applied to complex financial systems. To address these challenges, (Zhu et al., 2017) pioneered the network autoregressive (NAR) model, which effectively resolves the “curse of dimensionality” plaguing conventional VAR frameworks. Unlike traditional models where parameter counts grow exponentially with node numbers, the NAR specification maintains stable parameter dimensionality regardless of network size. This breakthrough enables robust analysis of ultra-high-dimensional financial time series data across multiple industries.

3. Data and Methodology

3.1. Data and sample description

We employ the indices based on the Shenwan Level-1 industry classification. The Shenwan industry classification system is built upon four core principles: “profit-driven, valuation clustering, physical attributes, and conventional industry practices” to establish its classification framework. The characteristics of these indices are as follows: they align with investors’ intuitive understanding, ensure that companies within the same industry share comparable business models, financial metrics, and market performance, and undergo timely adjustments in response to China’s economic development and the continuous evolution of the securities market, thereby ensuring well-grounded practicality of the industry definitions. The Shenwan Level-1 industry indices comprise 31 industries (Xu et al., 2022), the full names and abbreviations of which are provided in Appendix A (Table A1).

We adopt the period from January 2007 to December 2023 as the sample period. To analyze risk contagion patterns during distinct market regimes, we partition the sample into five periods based on major economic and financial events. Period A—August 2007 to December 2009—encompasses the global financial crisis, during which global credit markets experienced severe disruptions and contagion effects propagated across international markets (Brunnermeier, 2009; Longstaff, 2010). Period B—July 2014 to June 2015, and Period C—June 2015 to January 2016—capture China’s dramatic bull-to-bear market transition, characterized by leverage-fueled speculative bubbles (Bian et al., 2018; Allen et al., 2019). Period D—March 2018 to January 2020—covers the China-US trade friction episode, during which escalating tariffs disrupted supply chains and created uncertainty (Fajgelbaum et al., 2020; Fetzer and Schwarz, 2021). Period E—December

2019 to November 2020 — examines the COVID-19 pandemic’s impact, particularly the unprecedented market volatility (Baker et al., 2020; Ramelli and Wagner, 2020). These periods provide archetypal cases of entity-driven shocks (trade friction, pandemic supply disruptions) versus financial-driven shocks (leverage cycles, credit crunches), enabling systematic comparison of transmission mechanisms across shock types.

3.2. Research methodology

3.2.1. Multidimensional risk measurement method

To test our dual-track hypothesis—that entity-driven and financial-driven channels operate through distinct mechanisms—we must first quantify five risk dimensions that potentially capture different aspects of contagion: price risk (reflecting direct cost transmission), volatility risk (uncertainty propagation), asymmetric risk (downside contagion asymmetries), extreme risk (rare event spillovers), and tail risk (systemic leverage-driven contagion). Traditional single-dimensional approaches based solely on volatility or correlation cannot differentiate whether observed contagion reflects supply-chain mechanisms, financial panic, or both, thereby motivating our multidimensional measurement framework.

We employ the logarithmic returns of the 2021 edition of Shenwan Level-1 industry indices to measure price risk across industries. Specifically, the logarithmic return of industry indices is calculated as $R_t = \ln(P_t/P_{t-1})$, where P_t and P_{t-1} denote the closing prices of the industry indices at time t and $t - 1$, respectively.

For higher-moment risks, we apply the GARCHSK model proposed by León et al. (2005), using industry index returns to estimate conditional variance, conditional skewness, and conditional kurtosis, thereby measuring volatility risk, asymmetric risk, and extreme risk, respectively. Kurtosis risk reflects the cross-market propagation of extreme event risks, while skewness risk measures the transmission of return distribution asymmetries across markets (Bouri et al., 2021; Bouri and Jalkh, 2023). When the conditional variance, conditional skewness, and conditional kurtosis are all specified with GARCH (1, 1) structures, this model becomes the GARCHSK (1, 1; 1, 1; 1, 1) model, with the following expression:

$$\begin{cases} r_t = \mu_t + \varepsilon_t \\ \eta_t = h_t^{\frac{1}{2}} \varepsilon_t \\ h_t = \beta_0 + \beta_1 \eta_{t-1}^2 + \beta_2 h_{t-1} \\ s_t = \gamma_0 + \gamma_1 \eta_{t-1}^3 + \gamma_2 s_{t-1} \\ k_t = \delta_0 + \delta_1 \eta_{t-1}^4 + \delta_2 k_{t-1} \end{cases} \quad (1)$$

From top to bottom, these equations represent the mean equation, disturbance equation, variance equation, skewness equation, and kurtosis equation of the GARCHSK model, respectively. Here, r_t denotes the stationary series¹ of industrial stock index returns, μ_t represents its mean component, and the disturbance term ε_t satisfies $\varepsilon_t | I_{t-1} \sim GC(0, h_t, s_t, k_t)$, where I_{t-1} is the information set at time $t - 1$, and $GC(0, h_t, s_t, k_t)$ denotes the Gram-Charlier expanded normal distribution characterized by mean, variance, skewness, and kurtosis parameters. Under this

¹ Unit root tests conducted in both the static and dynamic analyses confirm that the logarithmic return series of the Shenwan Level-1 industry indices used in this study follow I(1) processes. In the empirical analysis, all series have been subjected to first-differencing to ensure stationarity requirements are met. The ADF test results for the logarithmic return series in the static analysis are provided in Appendix D Table A2; the ADF test results for the dynamic analysis are available upon request due to space constraints.

framework, the conditional variance $h_t = \text{var}_{t-1}(r_t) = \text{var}_{t-1}(\varepsilon_t)$, the conditional skewness $s_t = \text{skew}_{t-1}(r_t) = \text{skew}_{t-1}(\varepsilon_t)$, and the conditional kurtosis $k_t = \text{kurt}_{t-1}(r_t) = \text{kurt}_{t-1}(\varepsilon_t)$. The disturbance term is standardized using $h_t^{-1/2}$ to obtain a new disturbance term η_t , which satisfies $\eta_t | I_{t-1} \sim GC(0, 1, s_t^*, k_t^*)$, where s_t^* and k_t^* represent the standardized conditional skewness and kurtosis, respectively.

For tail risk measurement, we employ Conditional Value at Risk (CoVaR) following Pham et al. (2024). CoVaR quantifies systemic interdependence by measuring the maximum potential loss in one industry conditional on another industry experiencing extreme stress—a feature for identifying financial-driven contagion where leverage amplification creates nonlinear tail dependencies distinct from linear supply-chain transmission. We integrate CoVaR within our multidimensional framework rather than using it as a standalone measure, because isolated CoVaR analysis cannot distinguish whether tail dependencies stem from entity-driven linkages or financial-driven panic (Fengler and Gisler, 2015). It is essential to distinguish between fourth-moment-based extreme risk measurements (Bouri and Jalkh, 2023), which capture entity-specific rare event characteristics, and CoVaR-derived tail dependence risk (Pham et al., 2024), which quantifies systemic interdependence between individual industries and the overall market. The inherent conceptual divergence between these risk dimensions, as empirically demonstrated by the distinct temporal patterns revealed in our subsequent analyses, necessitates their coupled examination through an integrated framework. The detailed CoVaR estimation procedure using DCC-GARCH models (Engle, 2002) is provided in Appendix B.

To validate the multidimensional decomposition approach, we examine correlations among the five risk measures across all industries and time periods. Table 1 presents average pairwise correlations from rolling-window cross-sectional data. The correlation between price risk (log returns) and tail risk (CoVaR) is approximately zero (-0.007), indicating these dimensions capture fundamentally distinct risk characteristics. This near-orthogonality supports our dual-track hypothesis: if price risk and tail risk were highly correlated, entity-driven and financial-driven channels would be indistinguishable. Similarly, correlations between price risk and higher-order moments remain negligible ($|\rho| < 0.013$), while the highest absolute correlation across all pairs is only 0.146. This low-correlation structure justifies the decomposition approach as a methodological necessity rather than merely a technical choice.

Table 1

Average pairwise correlations among five risk dimensions

	price risk	tail risk	volatility risk	asymmetric risk	extreme risk
price risk	1.0000	-0.0069	0.0058	-0.0130	-0.0078
tail risk	-0.0069	1.0000	-0.0472	0.0060	0.0595
volatility risk	0.0058	-0.0472	1.0000	0.0546	-0.1461
asymmetric risk	-0.0130	0.0060	0.0546	1.0000	-0.0614
extreme risk	-0.0078	0.0595	-0.1461	-0.0614	1.0000

Note: This table reports element-wise mean correlations computed across all rolling windows (2007-2023) based on cross-sectional data of 31 industries. For each time window t , we construct a 31×5 matrix and compute the 5×5 correlation matrix. The near-zero correlation between price risk and tail risk ($\rho = -0.007$) validates that these dimensions capture distinct transmission mechanisms, supporting the dual-track hypothesis.

3.2.2. Multivariate nonlinear Granger causality test based on LOO algorithm and data sharpening

Having quantified five risk dimensions, we face the challenge of identifying causal transmission pathways for each dimension separately: which industries transmit price risk to others? Which propagate tail risk? Simply observing correlations cannot reveal directional causality, nor can linear methods capture the nonlinear amplification mechanisms inherent in financial contagion (fire sales, panic spillovers). Moreover, with 31 industries, spurious causality arising from third-party effects (e.g., industry A and B both respond to shocks in industry C, creating false direct linkage) becomes severe. We therefore require a methodology that (1) identifies nonlinear causal relationships, (2) controls for all potential third-party confounders, and (3) constructs dimension-specific causal networks to enable subsequent edge uniqueness analysis.

Therefore, we employ the multivariate nonlinear Granger causality test integrated with the leave-one-out (LOO) approach proposed by Hué et al. (2019) and enhanced with data-sharpening techniques by Diks and Wolski (2016). The definition of the multivariate Granger causality test is as follows. Given a strictly stationary multivariate time series $\{(X_t, Y_t, Q_t)\}$ for $t \in \mathbb{Z}$, where X_t and Y_t are univariate sequences, and Q_t is a univariate or multivariate sequence. For some $k \geq 1$, if:

$$(Y_{t+1}, \dots, Y_{t+k}) | (\mathcal{F}_{X,t}, \mathcal{F}_{Y,t}, \mathcal{F}_{Q,t}) \not\sim (Y_{t+1}, \dots, Y_{t+k}) | \mathcal{F}_{Y,t}, \mathcal{F}_{Q,t} \quad (2)$$

then $\{X_t\}$ is a Granger cause of $\{Y_t\}$. Here, $\mathcal{F}_{X,t}$, $\mathcal{F}_{Y,t}$, and $\mathcal{F}_{Q,t}$ represent the information contained in past observations X_s , Y_s , and Q_s (where $s \leq t$), respectively; the symbol “ $\sim$ ” denotes distributional non-equivalence.

To address the “curse of dimensionality” caused by excessive variables in vector autoregressive (VAR) systems, Hué et al., (2019) proposed a Granger causality test using the Leave-One-Out (LOO) approach to identify multivariate Granger causal relationships. The procedure is as follows:

Assume there are n stock market risk network nodes. Sequentially conditioning on node k ($k = 1, \dots, n$), perform conditional multivariate Granger causality tests to obtain $n \times n$ -dimensional network adjacency matrices W^{-k} ($k = 1, \dots, n$). In matrix W^{-k} , the k -th row and k -th column are set to 0, and the main diagonal elements are also set to 0. If nodes i and j exhibit a Granger causal relationship under the exclusion of node k , the element W_{ij}^{-k} is assigned 1; otherwise, it is assigned 0. Based on these n matrices, calculate the multidimensional sub-networks of stock market risk with direct Granger causal relationships between nodes:

$$W_{ij} = I \left(\sum_{k=1}^n W_{ij}^{-k} \right)_{[\sum_{k=1}^n W_{ij}^{-k} = n-2]} \quad (3)$$

Diks and Wolski (2016) extended the bivariate DP test (Phillips and Ouliaris, 1988) to multivariate scenarios by employing the data-sharpening method proposed by Hall and Minnotte (2002), which reduces the bias of kernel estimators. Consider strictly stationary time series $\{X_t\}$ and $\{Y_t\}$, where $\{Z_t\} = \{X_{t+1}\}$.

First, under the null hypothesis that $\{X_t\}$ is not a Granger cause of $\{Y_t\}$, the hypothesis is formulated as:

$$H_0: \quad \frac{f_{X,Y,Z}(x, y, z)}{f_{X,Y}(x, y)} = \frac{f_{Y,Z}(y, z)}{f_Y(y)} \quad (4)$$

which is equivalent to:

$$H'_0: \quad q \equiv E[f_{X,Y,Z}(x, y, z)f_Y(y) - f_{X,Y}(x, y)f_{Y,Z}(y, z)] = 0 \quad (5)$$

Second, the local kernel density estimator is defined as:

$$\hat{f}_W(W_i) = ((n-1)\varepsilon)^{-d_W} \sum_{j, j \neq i} K\left(\frac{W_i - W_j}{\varepsilon}\right) \quad (6)$$

where d_W denotes the dimension of matrix W , $i=1, \dots, n$, and ε represents the bandwidth for nonlinear Granger causality testing.

Third, following the approach of Choi and Hall (1999), a sharpened local kernel density estimator is constructed using a plug-in estimator based on the Nadaraya–Watson statistic:

$$\hat{f}_W^s(W_i) = ((n-1)\varepsilon)^{-d_W} \sum_{j, j \neq i} K\left(\frac{W_i - \psi_p(W_j)}{\varepsilon}\right) \quad (7)$$

with the sharpening operator $\psi_p(W)$ given by:

$$\psi_p(W) = I + \frac{\varepsilon^2 \sum_{i=1}^n (W_i - W) K\left(\frac{W_i - W}{\varepsilon_{DS}}\right)}{2\varepsilon_{DS}^2 K\left(\frac{W_i - W}{\varepsilon_{DS}}\right)} \quad (8)$$

Here, $\varepsilon_{DS} = C_{DS}n^{-1/(d_W+3)}$ is the data-sharpening bandwidth, where $C_{DS} = 4.0$ as calibrated by Diks and Wolski (2016).

Finally, the sharpened test statistic T_n^s is constructed as:

$$T_n^s(\varepsilon) = \frac{(n-1)}{n(n-2)} \sum_i (\hat{f}_{X,Y,Z}^s(X_i, Y_i, Z_i) \hat{f}_Y^s(Y_i) - \hat{f}_{X,Y}^s(X_i, Y_i) \hat{f}_{Y,Z}^s(Y_i, Z_i)) \quad (9)$$

The null hypothesis is rejected when the standardized statistic satisfies: $\sqrt{n}(T_n(\varepsilon_n) - q)/S_n \geq z_{1-\alpha}$, where S_n denotes the variance of T_n^s , estimated using the method proposed by Diks and Panchenko (2006).

3.2.3. Multidimensional network coupling method

The five dimension-specific causal networks constructed above capture distinct transmission mechanisms, but analyzing them in isolation obscures a key question: how does the relative importance of each dimension shift across time? During the 2015 crash, did tail risk suddenly dominate, or did all dimensions intensify proportionally? Answering this requires synthesizing information across all five networks to estimate time-varying weights that quantify each dimension's contribution to aggregate systemic risk—the key evidence for demonstrating that the financial-driven channel (tail risk) becomes explosively dominant during crises while the entity-driven channel (price risk) maintains stable persistence.

We employ the multidimensional network coupling approach (Bonaccolto et al., 2019; Billio et al., 2023) which extends spatial econometric models to multilayer network settings. This approach yields two outputs: (1) the coupled network adjacency matrix $W^* = \sum_{d=1}^D \delta_d W_d$, and (2) time-varying coupling weights $\delta_d(t)$ that enable us to track how the dominant driver of systemic risk switches from entity-driven to financial-driven channels during crises. The detailed coupling procedure is provided in Appendix C.

3.2.4. Network Topological Property indices

We examine the topological characteristics of stock market risk contagion networks across three hierarchical levels, for an overall understanding of the networks constructed. At the whole-network level, network density ($Dens$) is utilized to evaluate the sparsity of connections, calculated as the ratio of observed links to the maximum possible links in the network. A higher network density indicates more extensive risk transmission channels and accelerated risk contagion across the system.

At the local structure level, the average clustering coefficient (ACC) quantifies the density of community structures¹, reflecting the intensity of community formation within the network. An elevated ACC suggests stronger local clustering effects, where neighboring nodes exhibit dense interconnectivity. This structural property implies that localized risk containment strategies targeting individual nodes may fail to mitigate systemic contagion. The ACC is formally defined as:

$$ACC = \frac{1}{n} \sum_{i=1}^n ACC_i = \frac{1}{nd_i(d_i - 1)} \sum_{j,k=1}^n \frac{W_{ij} + W_{ik}}{2\langle W_i \rangle} a_{ij}a_{ik}a_{jk} \quad (10)$$

At the individual node level, the average eigenvector centrality (EC) quantifies the systemic influence of the nodes. A higher average EC suggests that dominant industries act as hubs for risk contagion, where localized shocks may cascade to the whole market (Yu et al., 2024). The eigenvector centrality vector $\mathbf{EC} = (EC_1, EC_2, \dots, EC_n)$ is derived from the spectral decomposition of the adjacency matrix, corresponding to the eigenvector associated with the largest eigenvalue λ . For node i (where W_{ij} is the edge weight between nodes i and j):

$$EC_i = \frac{1}{\lambda} \sum_{j=1}^n W_{ij} EC_j \quad (11)$$

3.2.5. Network configuration analysis methods

To empirically disentangle the dual tracks, we require metrics that measure: (1) Edge Uniqueness (EU): the extent to which each dimension possesses distinct contagion paths unexplained by other dimensions—our primary test of whether price risk and tail risk operate through fundamentally different mechanisms. (2) Block structure: how industries cluster into risk transmission communities and whether these clusters differ across dimensions. (3) Node systemic importance: which industries serve as central propagators versus absorbers in each dimension-specific network, enabling identification of dimension-specific hubs.

(1) Dimensional heterogeneity analysis in multidimensional networks

Edge Uniqueness (EU) measures the distinctiveness of a specific dimension relative to other dimensions in a multidimensional network. It quantifies the proportion of edges that exist in a given dimension but are absent in all other dimensions, thereby assessing the unique contribution of that dimension to the multidimensional network (Musmeci et al., 2017). For a multidimensional asymmetric network, we define the Edge Uniqueness of dimension α as:

¹ In this study, the industry-level community structure is termed “industry-specific communities”, which are characterized by significantly stronger risk contagion linkages among industries within the same community, while risk transmission intensities between distinct communities are comparatively weaker.

$$EU^{\{\alpha\}} = \frac{1}{TL^{\{\alpha\}}} \sum_{i \neq j} \left[W_{ij}^{\{\alpha\}} \prod_{\beta \neq \alpha} (1 - W_{ij}^{\{\beta\}}) \right] \quad (12)$$

where $TL^{\{\alpha\}} = \sum_{i \neq j} W_{ij}^{\{\alpha\}}$ represents the total number of edges in dimension α ; $W_{ij}^{\{\alpha\}}$ denotes the adjacency relationship from node i to node j in dimension α (taking value 1 if the edge exists and 0 otherwise); β iterates over all dimensions except α ; $\prod_{\beta \neq \alpha} (1 - W_{ij}^{\{\beta\}})$ equals 1 only when the edge (i, j) is absent in all other dimensions. The Edge Uniqueness ranges between $[0, 1]$, with higher values indicating more distinctive structural features of the dimension and less overlap with other dimensions.

(2) Block model analysis

The block model is a framework for describing the community structure of networks, which divides nodes into groups based on their attributes or behavioral rules, forming several “communities” or “blocks” within a complex network. This method not only identifies the nodes contained in each block but also analyzes the role each block plays in a risk contagion network (Zhang et al., 2020). The block model classifies network blocks into four distinct roles: First, the Main Benefit, where members receive both internal and external links but obtain more links within the block than from outside, maintaining a small proportion of external relations. Second, the Main Spillover, characterized by members that send many links to other blocks while exchanging few links internally and receiving minimal external connections. Third, the Bilateral Spillover, whose members actively send numerous links both inside and outside their blocks while receiving few external links in return. Finally, the Broker, where members both send and receive external links while maintaining few internal connections.

Our block model analysis account for how a block’s relative size influences its internal relationship count, where larger blocks inherently contain more internal connections, whereas smaller blocks demonstrate the inverse pattern (Wasserman and Faust, 1994). To accurately evaluate a block’s relationship preferences, one should compare the actual proportion of internal relationships against the expected proportion, while accounting for both the possible relationships within the block and across the entire network.

For example, when analyzing the relationships among members in the block B_m , if B_m contains n_m nodes, then the maximum possible number of internal relationships in B_m is $n_m(n_m - 1)$. If the entire network consists of n nodes, then the total possible number of relationships for members in B_m is $n_m(n - 1)$. Thus, the expected proportion of internal relationships in the block B_m relative to its total relationships is:

$$\frac{n_m(n_m - 1)}{n_m(n - 1)} = \frac{n_m - 1}{n - 1} \quad (13)$$

By further incorporating the proportion of external relationships received by the block members, the blocks can be classified into four distinct role types (see Table 2).

Table 2

Classification criteria for blocks in the block model.

Internal Linkages Ratio	Received Linkages Ratio
-------------------------	-------------------------

	≈ 0	> 0
$\geq (n_m - 1)/(n - 1)$	Bilateral Spillover	Main Benefit
$< (n_m - 1)/(n - 1)$	Main Spillover	Broker

Note: This table presents the classification criteria for block roles in the block model. Based on the comparison between a block's actual and expected proportion of internal relationships, as well as the proportion of external relationships received by its members, blocks are categorized into four distinct roles: Main Benefit, Main Spillover, Bilateral Spillover, and Broker.

(3) Node Systemic Importance Measurement

The System Importance Measure (SIM) for network nodes is calculated using the following procedure. Consider a multidimensional financial risk contagion network with n nodes. First, define the total number of linkages in the network as:

$$TL = \sum_{i \neq j} W_{ij} \quad (14)$$

Subsequently, when node k is removed, the remaining sub-network comprising $n-1$ nodes exhibits a linkage count of:

$$TL^{-k} = \sum_{i \neq j} W_{ij}^{-k} \quad (15)$$

The systemic importance of node k is then quantified as:

$$SIM_k = \frac{|TL - TL^{-k}|}{TL} \quad (16)$$

This metric captures the relative disruption to network connectivity caused by the removal of node k , normalized by the original network's total linkages. The absolute value operator ensures directional neutrality in assessing systemic impact.

4. Results and analysis

4.1. Dynamic analysis of multidimensional risk contagion networks

Using time-series data of five-dimensional risks across industries, we construct dynamic stock market risk networks through nonlinear multivariate Granger causality tests based on the leave-one-out approach. Following established practices in the Chinese market literature, we employ a rolling window of 250 trading days (approximately one trading year) with a sliding step of 5 trading days, yielding weekly frequency networks from 2007 to 2023. This framework generates five dimension-specific causal networks—price risk, volatility risk, asymmetric risk, extreme risk, and tail risk networks—alongside a synthesized multidimensional coupled network that integrates information across all dimensions. The analysis proceeds in three stages: first, we examine when systemic stress emerges through aggregate network topology; second, we disentangle why stress emerges by isolating dimension-specific transmission mechanisms via edge uniqueness analysis; third, we validate these mechanisms through sectoral contribution patterns.

4.1.1. Coupled network topology evolution

Fig. 1 illustrates the temporal evolution of these three metrics. Several patterns emerge from the data. First, all three metrics exhibit synchronized fluctuations, suggesting that changes in global

connectivity, local clustering, and node centrality reflect common underlying forces rather than independent structural shifts. Second, during exogenous shock periods—notably the global financial crisis and the COVID-19 pandemic—the network demonstrates what we term a “triple-high” configuration. Network density rises substantially, indicating proliferation of cross-sector linkages; clustering coefficients increase, signaling strengthened community structures; and eigenvector centrality becomes more concentrated, reflecting the dominance of systemically important nodes. These patterns are consistent with theoretical predictions that external shocks propagate risks through multiple channels—capital flow disruptions, supply chain fractures, and sentiment contagion—amplified by leverage effects and policy intervention lags (Chan, 2017; Yu et al., 2024).

Third, during endogenous bull-to-bear transitions (Period B: July 2014 to June 2015; Period C: June 2015 to January 2016), the metrics display sharp “V-shaped” reversals. All three indicators decline rapidly during the bull market phase, then surge explosively following the market crash. This pattern suggests that leverage-driven risks remained concealed under market euphoria, only to erupt when sentiment reversed or deleveraging policies triggered liquidity crunches. Notably, while risk metrics moderated during the post-2015 regulatory reforms and the China-US trade friction period (Period D: March 2018 to January 2020), they did not return to pre-crisis baselines, indicating persistent structural vulnerabilities.

These topological patterns document when the risk network intensifies but cannot explain the underlying mechanisms. The coupled network aggregates all five risk dimensions, obscuring whether intensification reflects supply-chain disruptions propagating price shocks, financial panic amplifying tail dependencies, or both operating simultaneously. This limitation motivates our decomposition analysis in the following subsection.

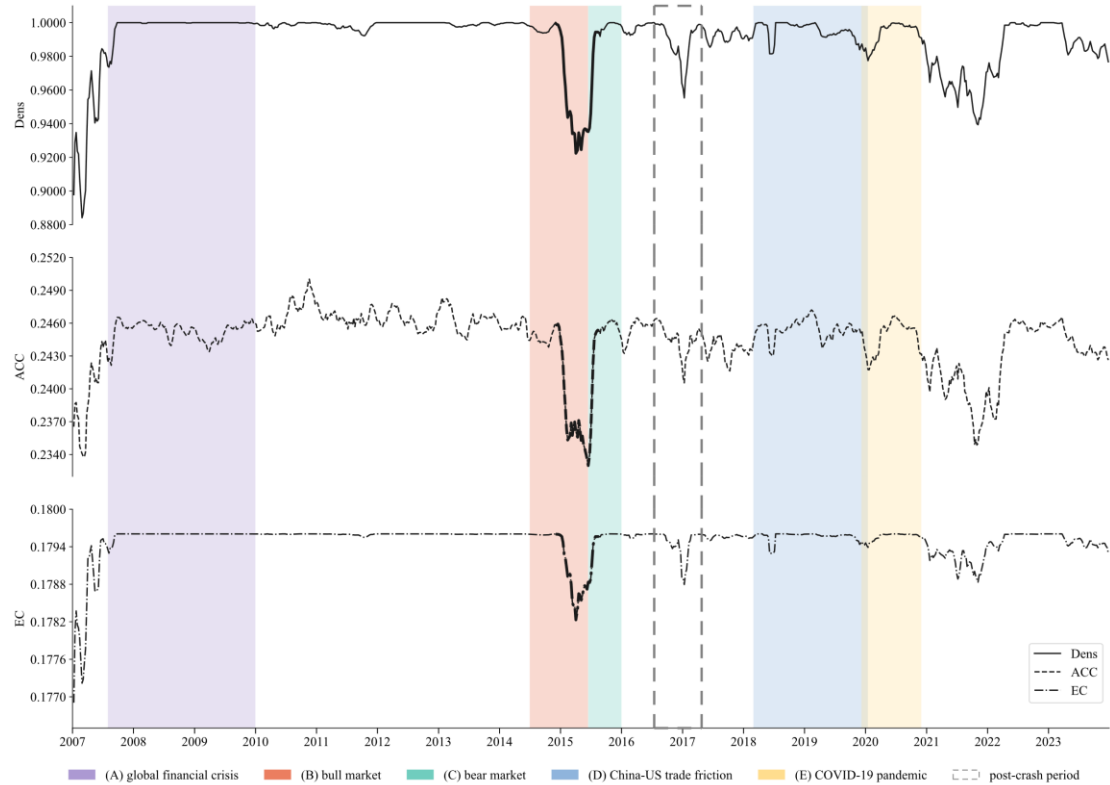


Fig. 1. Temporal evolution of topological properties in the five-dimensional risk contagion coupled network.

Note: This figure charts network density (Dens, solid line), average clustering coefficient (ACC, dashed line), and eigenvector centrality (EC, dash-dot line) from 2007 to 2023. Shaded regions mark: (A) global financial crisis, (B) bull market, (C) bear market, (D) China-US trade friction, (E) COVID-19 pandemic. The grey dashed rectangle indicates the 2016-2017 “post-crash period”. Time series smoothed via 5-period moving average.

4.1.2. Dimension-specific transmission mechanisms

Having established aggregate temporal patterns, we now disentangle heterogeneous transmission mechanisms by examining edge uniqueness across the five risk dimensions. Edge uniqueness quantifies the extent to which risk transmission pathways in one dimension are distinct from those in other dimensions (Musmeci et al., 2017). Higher values indicate that a specific risk dimension possesses contagion channels unexplained by other metrics, thereby offering incremental information about systemic risk propagation. This analysis constitutes the core test of our dual-track hypothesis: if entity-driven and financial-driven channels operate through fundamentally distinct mechanisms, we should observe systematically different edge uniqueness patterns for price risk (our proxy for supply-chain transmission) versus tail risk (our proxy for financial-channel contagion).

Fig. 2 presents the dynamic evolution of edge uniqueness for all five risk dimensions from 2007 to 2023. The results reveal striking heterogeneity. Price risk networks consistently maintain elevated edge uniqueness throughout the sample period, with values generally exceeding those of other dimensions except during extreme market turmoil. This persistence suggests that cost and demand transmission along supply chains generates contagion pathways that remain structurally distinct from those captured by volatility, skewness, or kurtosis metrics, regardless of market regime. The stability of price risk uniqueness reflects the continuous nature of real-economy production networks: industries remain interconnected through input-output linkages that propagate price shocks via cost transmission mechanisms, even when financial markets transition between boom and crash cycles.

In sharp contrast, tail risk networks exhibit dramatically different temporal dynamics. During tranquil periods—including most of the pre-crisis years, the bull market phase, and the China-US trade friction period—tail risk uniqueness remains subdued, often ranking among the lowest across all five dimensions. This dormancy indicates that tail dependencies largely overlap with patterns captured by other risk metrics during normal market conditions. However, during another three episodes—the bear market phase, segments of the COVID-19 pandemic, and portions of the global financial crisis—tail risk uniqueness surges to levels comparable to or exceeding price risk uniqueness. These eruptions signal activation of a crisis-specific transmission mechanism: systemic tail dependencies driven by leverage unwinding, liquidity freezes, and panic-induced correlation breakdowns that create contagion pathways qualitatively different from gradual supply-chain adjustments.

The remaining three dimensions—volatility risk, asymmetric risk, and extreme risk—generally exhibit moderate edge uniqueness with limited temporal variation. Their contagion pathways largely overlap with those of price and tail risk, suggesting these dimensions capture intermediate or composite aspects of transmission rather than constituting independent channels. Notably, extreme risk (kurtosis-based) displays modest increases during crises but remains substantially below tail

risk peaks, confirming that fourth-moment measures of rare events differ conceptually from CoVaR-derived systemic tail dependencies that condition on joint distress.

These patterns provide direct empirical support for the dual-track hypothesis. The persistent elevation of price risk uniqueness, coupled with its relative stability across regimes, indicates that entity-driven transmission through supply chains operates continuously as a structural feature of inter-industry risk propagation. Conversely, the crisis-specific eruptions of tail risk uniqueness demonstrate that financial-driven transmission activates only under extreme stress, becoming the dominant channel when leverage amplification and panic contagion override gradual cost transmission mechanisms.

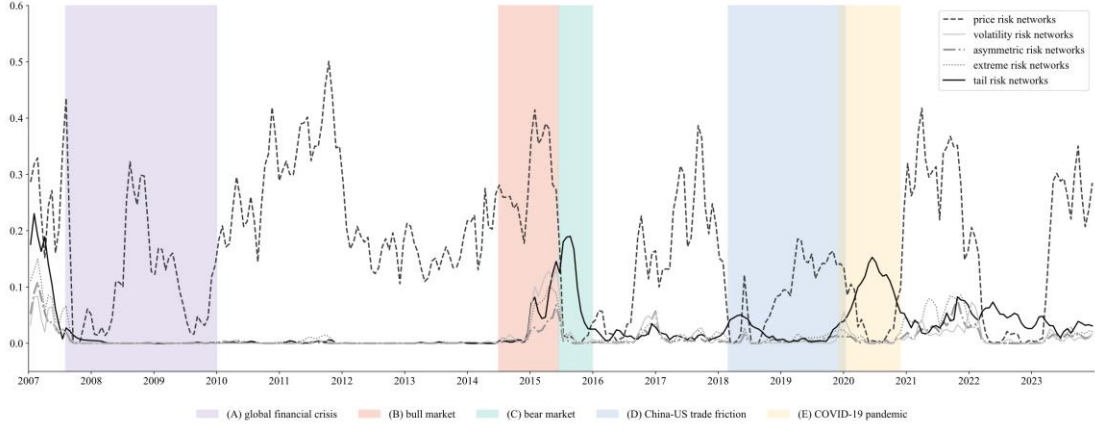


Fig. 2. Dynamic evolution of edge uniqueness across five-dimensional risk contagion subnetworks.

Note: This figure illustrates edge uniqueness for price risk (dashed line), volatility risk (thin solid line), asymmetric risk (dash-dot line), extreme risk (dotted line), and tail risk (thick solid line) from 2007 to 2023. Higher values indicate more unique contagion paths offering greater incremental informational value. Shaded regions mark periods A-E as in Fig. 1. Time series smoothed via 5-period moving average.

To provide additional quantitative evidence for the dual-track mechanism, we examine the temporal evolution of subnetwork coupling weights estimated via the optimization procedure described in Section 3.2.3. These time-varying weights quantify each risk dimension’s relative contribution to aggregate systemic risk at time t , measuring which dimension dominates in driving overall network connectivity. This analysis complements the edge uniqueness results in Fig. 2. Edge uniqueness measures pathway distinctiveness—whether a dimension possesses unique transmission channels. Coupling weights measure dominance—whether a dimension drives aggregate risk. Together, these metrics provide convergent validation of the dual-track mechanism.

Fig. 3 presents the dynamic evolution of coupling weights for all five risk dimensions from 2007 to 2023. Price risk (log returns) maintains remarkably stable weights throughout the entire sample period. It consistently hovers around 0.20 with minimal fluctuation across all market regimes, including the 2008 global financial crisis, the 2015 crash, and the 2020 pandemic. This stability pattern aligns consistently with the edge uniqueness evidence. Fig. 2 showed that price risk networks maintain persistently high uniqueness across all periods. Fig. 3 now demonstrates that this unique transmission pathway also contributes steadily to aggregate risk. The convergence of high uniqueness and stable moderate weights confirms that entity-driven transmission through supply

chains operates continuously. It possesses both distinct pathways and consistent importance regardless of market conditions. This dual confirmation validates the structural, always-present nature of the entity-driven channel.

Tail risk (CoVaR) exhibits a fundamentally different temporal profile. Throughout most of the sample period, tail risk coupling weights fluctuate around levels comparable to other dimensions, which remain largely indistinguishable from volatility or asymmetric risk contributions. However, during the 2015 stock market crash, tail risk weights surge dramatically to their sample-period maximum, substantially exceeding all other dimensions. This weight pattern reveals important heterogeneity when compared with edge uniqueness evidence—during 2015, both metrics peaked simultaneously. Fig. 2 showed tail risk uniqueness surging to sample highs, and Fig. 3 shows tail risk weights dominating aggregate connectivity. This convergence of high uniqueness and high dominance indicates complete activation of the financial-driven channel: leverage unwinding created both distinct transmission pathways and overwhelming contribution to systemic risk. In contrast, during 2020, a critical divergence emerges: tail risk uniqueness surged (Fig. 2) while coupling weights remained moderate (Fig. 3). This divergence suggests that although COVID-19 activated tail dependencies creating unique contagion pathways, these pathways did not dominate aggregate risk. The pattern is consistent with the pandemic representing primarily a supply-chain shock rather than a systemic leverage crisis.

The remaining three dimensions—volatility, asymmetric, and extreme risk—display substantial temporal variation in coupling weights across different periods. These fluctuations reflect that higher-order moment risks capture intermediate or composite aspects of transmission, whose relative contributions shift depending on shock characteristics. Comparing these dimensions' coupling weights with their edge uniqueness (Fig. 2) clarifies an essential distinction: moderate uniqueness combined with variable weights indicates dimensions that possess partially distinct pathways; however, their importance fluctuates with market conditions. This pattern differentiates them from the two primary channels: the entity-driven channel exhibits stable weights and persistent uniqueness, while the financial-driven channel exhibits explosive weights during specific episodes and episodic uniqueness surges. The 2015 episode thus stands as the definitive validation, during which tail risk simultaneously achieved maximum uniqueness and maximum coupling weights. This demonstrates that financial-driven contagion can completely dominate systemic risk when leverage amplification activates. Such dominance represents a phenomenon fundamentally distinct from the continuous, moderate contribution of entity-driven transmission.

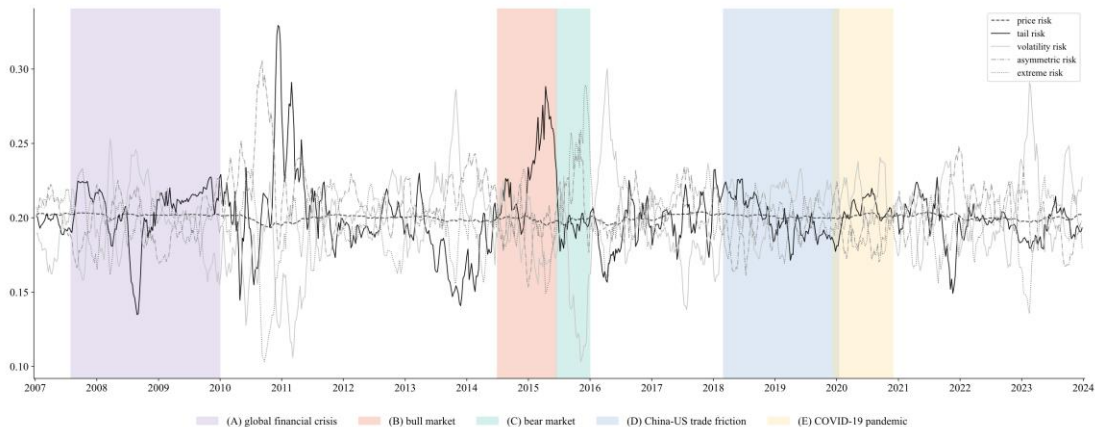


Fig. 3. Temporal evolution of risk dimension coupling weights.

Note: This figure presents the time-varying coupling weights for the five risk dimensions from 2007 to 2023. Higher weights indicate greater contribution to aggregate systemic risk. Shaded regions mark periods A-E as in Fig. 1. Time series smoothed via 5-period moving average.

4.1.3 Sectoral contribution patterns

To validate the dual-track mechanism identified through edge uniqueness analysis, we examine whether sectoral risk contributions align with the predicted patterns. If price risk reflects continuous supply-chain transmission, sectors deeply embedded in production networks should contribute persistently regardless of market regime. Conversely, if tail risk activates during crises, the financial sector's contribution should surge precisely when tail risk uniqueness peaks.

We compute dynamically weighted average systemic importance measures for five major sector groups: Financial (banking and non-bank finance), Technology (electronics, computers, communications, media), Cyclical (oil & petrochemical, coal, steel, nonferrous metals, building materials, machinery, real estate, transportation), Consumption (food & beverage, textiles, household appliances, pharmaceuticals, agriculture), and Comprehensive (defense, utilities, public services). The 31 industries are categorized based on their positions in industrial chains. For each sector group, we calculate the weighted average of constituent industries' systemic importance measures, using time-varying coupling weights to aggregate across risk dimensions. The relative contribution of each sector at time is computed as its weighted systemic importance divided by the sum across all five sectors.

Fig. 4 presents the temporal evolution of sectoral risk contributions from 2007 to 2023. The results support the dual-track interpretation. First, the Cyclical sector maintains the highest and most stable contribution throughout nearly the entire sample period, generally ranging between 25% and 35%. This stability aligns with our interpretation of price risk as reflecting continuous supply-chain transmission: cyclical industries form the backbone of China's production networks, transmitting cost and demand shocks persistently through upstream-downstream linkages. The sector's contribution exhibits only moderate increases during the global financial crisis and COVID-19 pandemic, consistent with the relative stability of price risk uniqueness during these periods. Small fluctuations occur but do not fundamentally alter the sector's dominant position, confirming that entity-driven transmission constitutes a structural, always-present channel.

Second, the Financial sector's contribution displays explosive crisis-specific surges that coincide precisely with tail risk uniqueness peaks. During the global financial crisis, the Financial sector's contribution jumps substantially, driven by liquidity crunches in banking and leverage collapses in non-bank finance. Even more dramatically, during the bear market phase following the 2015 bull market peak, the Financial sector's contribution surges to levels exceeding 40%, dominating the risk transmission network as off-market margin financing unwound and stock pledge financing triggered forced liquidations. These eruptions occur precisely when tail risk uniqueness surges in Fig. 2, providing sectoral validation that financial-driven contagion becomes overwhelmingly dominant when systemic tail dependencies activate. Notably, during the China-US trade friction period—an entity-driven shock—the Financial sector's contribution remains subdued

despite elevated overall risk, confirming that the financial channel remains dormant when crises do not involve leverage amplification or panic contagion.

Third, the Technology sector exhibits an intriguing dual pattern: elevated contributions during both the 2015 crash and the trade friction period, but through different mechanisms suggested by the timing and co-movement with other sectors. During the crash, the Technology sector's surge coincides with the Financial sector's peak and tail risk activation, suggesting vulnerability through high valuations and speculative leverage—a financial-channel effect. During trade friction, the Technology sector's elevated contribution occurs while tail risk uniqueness remains low and coincides with stable Cyclical sector contributions, indicating impact through export restrictions and supply chain restructuring—an entity-channel effect. This dual exposure demonstrates that some sectors participate in both transmission channels, though the mechanisms differ fundamentally.

The Consumption sector maintains relatively stable contributions for most periods but exhibits increases during the COVID-19 pandemic, reflecting supply chain disruptions for essential goods. The Comprehensive sector displays delayed responses, with contributions rising after other sectors have peaked, potentially reflecting its role in absorbing residual risks through diversified industrial exposures.

Taken together, the sectoral evidence corroborates the dual-track mechanism identified through edge uniqueness analysis. The Cyclical sector's persistent dominance parallels price risk's continuous elevation, while the Financial sector's crisis-specific surges align with tail risk activations. This correspondence between dimension-specific edge uniqueness patterns and sectoral contribution dynamics provides convergent validation that entity-driven and financial-driven channels operate through fundamentally distinct mechanisms with different temporal characteristics.

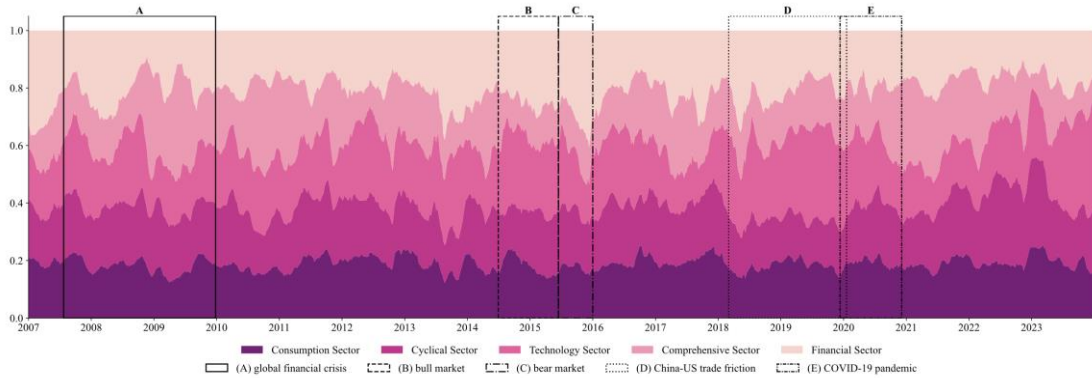


Fig. 4. Temporal changes in risk contagion contributions of the five major sectors.

Note: This figure presents a stacked area chart illustrating the time-varying proportional contributions of five major economic sectors to the overall risk in the multidimensional stock market contagion network from 2007 to 2023. The sectors shown are consumption, cyclical, technology, comprehensive, and financial. Each sector's contribution is calculated from a systemic importance measure, which represents a weighted average of industry-level risk aggregated based on coupling weights from each risk contagion subnetwork. The dashed boxes highlight key historical intervals: (A) the global financial crisis, (B) the bull stock market, (C) the bear stock market, (D) the China-US trade friction, and (E) the COVID-19 pandemic. All the data series have been smoothed using a 5-period moving average.

4.2. Static analysis of multidimensional risk contagion networks

The dynamic analysis provides temporal evidence that the multidimensional risk network exhibits patterns consistent with a dual-track contagion mechanism. To substantiate these dynamic findings and elucidate the underlying transmission mechanisms at the sectoral level, we now examine static network configurations during five historical periods. Rather than exhaustively documenting all periods, we organize our analysis around two contrasting case studies that represent fundamentally different shock origins. Throughout the following analysis, we employ standard abbreviations for industry names (e.g., ET for Electronics, NBF for Non-Bank Finance, CP for Computer) to enhance readability. The complete list of full names and abbreviations is provided in Appendix A (Table A1).

4.2.1. Case 1: China-US trade friction—An entity-driven channel shock

The China-US trade friction period (March 2018 – January 2020) constitutes a typical entity-driven channel shock. Trade policy disruptions directly impacted supply chains, production costs, and export-oriented sectors—mechanisms characteristic of entity-driven contagion.

Following Zhang et al. (2020), we conduct block model analysis on the constructed five-dimensional risk contagion networks, with inter-block interaction patterns and block classifications presented in Table 3. During the China-US trade friction, M1 and M2 were both classified as main benefit blocks. M1 included all technology sector industries (electronics, computer, communications, media), cyclical sector industries such as transportation and real estate, and consumption sector industries like pharmaceuticals and biotechnology, and food and beverage. The underlying mechanism lies in the direct shocks inflicted upon the technology sector during China-US trade friction through technology embargoes, export restrictions, and supply chain restructuring, which significantly intensified its risk absorption capacity and positioned it as a risk absorption conduit (Zhou et al., 2025). For instance, core technological domains such as semiconductors and communication equipment were subjected to external policy suppression during this period. This not only concentrated risks internally within these industries but also heightened their vulnerability, leading to the absorption of additional spillover risks from other sectors.

M2 primarily comprised some consumption and cyclical industries alongside the non-bank finance industry. These industries absorbed risks from other blocks during the trade friction phase due to external shocks. For instance, the nonferrous metals and chemical industries faced impacts from trade policy uncertainty and global supply chain volatility, while the non-bank finance industry became a primary absorber of external risks due to funding chain fluctuations. M3, classified as the main spillover block, included cyclical sectors such as machinery equipment and automobile industries, the banking industry, and the comprehensive industry. The machinery and automobile industries exhibited significant risk spillovers under tariff pressures and export dependencies, while the banking industry intensified risk transmission capabilities amid exchange rate volatility and liquidity tightening. M4, designated as the bilateral spillover block, encompassed oil and petrochemical, architectural ornament, electrical equipment, national defense, and steel industries. The oil and petrochemical and steel industries functioned as both risk sources and transmitters under dual pressures of global trade dynamics and commodity price fluctuations. Concurrently, the

national defense industry demonstrated strong bidirectional risk interactions, reflecting heightened geopolitical tensions between China and the U.S. during this period.

The image matrix for Period D (Fig. 5) reveals a unidirectional transmission pattern characteristic of supply chain propagation. Block 4 (bilateral spillover, containing upstream manufacturing) transmits risks predominantly toward Block 1 (main benefit, containing technology sectors). Block 2 (main benefit) receives inputs from Block 4 but exhibits minimal outward transmission. This unidirectional flow from upstream manufacturing to downstream absorption sectors validates the supply chain logic underlying entity-driven contagion.

Building on the block-level analysis, we examine individual industry roles within the multidimensional risk contagion networks (see Fig. 6). Industries are classified into four roles based on their in-degree and out-degree rankings: Risk Drivers (low in-degree, high out-degree), Risk Recipients (high in-degree, low out-degree), Risk Propagators (high in-degree, high out-degree), and Risk Isolates (low in-degree, low out-degree). Fig. 6 Panel D maps these roles across the China-US trade friction period. The four technology industries—ET (Electronics), CP (Computer), CMC (Communications), and MD (Media)—are all concentrated in the right half of the diagram, indicating high in-degree rankings (absorbing substantial risks from other sectors). Specifically, ET, CP, CMC, and MD occupy positions with out-degree rankings between 0-5 and in-degree rankings between 5-15, placing them predominantly in the Risk Recipients quadrant. This cohesive clustering confirms that technology industries collectively functioned as risk absorbers during the trade friction.

In contrast, upstream manufacturing industries occupy distinctly different positions. TA (Textile and Apparel), AM (Automobile Manufacturing), and AFAHF (Agriculture, Forestry, Animal Husbandry and Fishery) locate in the upper-left quadrant (low in-degree, low-to-moderate out-degree), functioning as Risk Drivers. Meanwhile, the non-bank financial industry (NBF) and banking industry (BA) occupy moderate positions without exhibiting dominant transmission roles—NBF has an in-degree ranking around 15 and out-degree ranking around 24, while BA sits at approximately in-degree 10 and out-degree 11. The systemic importance measures, indicated by node sizes in Fig. 6 Panel D, reveal that consumption and cyclical sectors (shown in red and orange) such as SS (Steel), FB (Food and Beverage), BC (Building and Construction), and HA (Household Appliances) exhibit relatively large node sizes, reflecting their centrality in production networks. Notably, the largest nodes are AFAHF, OP (Oil and Petrochemical), FB, and SL (Steel), all cyclical or consumption sectors deeply embedded in supply chains. In contrast, NBF and BA display moderate node sizes, indicating that leverage-based importance remained subdued when financial channels were dormant.

These static patterns align with the finding that tail risk edge uniqueness remained subdued during Period D (Fig. 2). The concentration of technology industries in risk absorption roles, the unidirectional transmission from upstream manufacturing sectors, and the peripheral positioning of financial institutions collectively validate that the China-US trade friction operated primarily through the entity-driven channel, where risks propagated along supply chain linkages rather than through financial intermediation.

4.2.2. Case 2: 2015 stock market crash—A financial-driven channel shock

The 2015 stock market crash (June 2015 – January 2016) represents a typical financial-driven channel shock. Triggered by aggressive deleveraging policies that unwound margin financing, this

crisis emanated from leverage concentration rather than real economic fundamentals—mechanisms characteristic of financial-driven contagion.

Table 3 Period C documents a marked structural transformation from the trade friction period. Crucially, the four technology industries exhibited fragmented positioning: Computer (CP) in M2, while Electronics (ET), Communications (CMC), and Media (MD) all in M3. This dispersion contrasts sharply with their cohesive clustering in M1 during Period D, suggesting that financial panic disrupted sectoral coherence. Most notably, the non-bank financial industry (NBF) transitioned from M2 (main benefit) during trade friction to M4 (main spillover) during the crash, signaling its emergence as the primary source of systemic risk transmission. When leverage unwinding drives contagion, risks propagate through financial relationships—cross-holdings, margin calls, liquidity spirals—that cut across industrial boundaries, severing the supply chain logic that unified technology sectors under entity-driven shocks.

The image matrix for Period C (Fig. 5) reveals a fundamentally different transmission topology. Unlike Period D's unidirectional flows from upstream manufacturing to downstream absorption, Period C exhibits bidirectional amplification dynamics. Block 2 (main benefit) displays a prominent self-loop, indicating intense internal risk circulation among its members. Block 4 (main spillover, containing NBF) transmits risks upward toward Block 1 (main benefit), while Block 3 (main spillover, containing technology sectors) sends spillovers laterally to Block 2. Critically, Block 3 receives flows from Block 2, creating a bidirectional exchange absent in the supply chain-driven Period D topology. This mutual amplification between blocks validates that risks propagated through financial linkages rather than production networks.

Fig. 6 Panel C demonstrates how individual industries repositioned during the crash. The technology sector's fragmentation becomes visually apparent: Electronics (ET) locates at in-degree 24 and out-degree 5, Computer (CP) at in-degree 3 and out-degree 11, Communications (CMC) at in-degree 26 and out-degree 15, and Media (MD) at in-degree 25 and out-degree 16. This scattered distribution across different quadrants—with CP shifting rightward and upward while ET, CMC, MD remain dispersed in the left-center regions—contrasts starkly with their tight clustering in Period D's right half, confirming that leverage exposure rather than supply chain position determined individual industries' vulnerability during financial crises.

These static patterns align precisely with the documentation of tail risk edge uniqueness surging dramatically during Period C (Fig. 2). The explosive activation of tail risk networks, combined with NBF's transition from main benefit to main spillover, the bidirectional amplification flows in Fig. 5 Panel C (contrasting with unidirectional supply chain flows in Fig. 5 Panel D), and the fragmentation of previously cohesive technology sectors collectively validate that the 2015 crash operated predominantly through the financial-driven channel. Unlike trade friction's supply chain propagation, the crash transmitted via leverage pyramids, liquidity spirals, and sentiment contagion—mechanisms captured by tail risk but not price risk networks.

To sum up, the contrasts between Periods C and D substantiate the dual-track hypothesis through four dimensions. First, NBF's role reversed from a main benefit block during trade friction to a main spillover block during the crash. This demonstrates that financial institutions' function fundamentally shifts between entity-driven and financial-driven regimes. Second, technology sectors transformed from cohesive clustering in M1 during Period D to fragmentation across M2

and M3 during Period C. Their tight concentration in the right half of Fig. 6 Panel D contrasts with dispersion across quadrants in Fig. 6 Panel C, which reveals that supply chain coherence collapses under leverage-driven contagion. Third, the transmission topology inverted from unidirectional flows to bidirectional amplification loops. Fig. 5 Panel D shows Block 4 transmitting toward Block 1, while Fig. 5 Panel C displays Block 2's self-loop and mutual exchanges between Blocks 2 and 3. This confirms that entity-driven channels follow production networks, while financial-driven channels operate through self-reinforcing panic dynamics. Fourth, these static findings triangulate with the dynamic evidence, including the persistent high uniqueness of price risk networks versus explosive surges in tail risk networks, and the stable cyclical contributions versus crisis-period financial spikes. This convergence across multiple analytical approaches establishes that China's inter-industry risk contagion operates through two coexisting yet distinct channels, each activated by different shock origins and transmitting through fundamentally different mechanisms.

Table 3

Interaction patterns and block classification results.

Blocks	Receiving relationship				Number of members	Expected internal relation ratio (%)	Actual internal relation ratio (%)	Receive links from outside	Emit links to outside	Block Roles	Industry Nodes
	M1	M2	M3	M4							
Period A: the global financial crisis											
M1	40	44	28	52	8	23.33	24.39	67	124	Bilateral Spillover	TP, PB, PU, ET, CM, AM, ND, LM
M2	34	64	46	63	11	33.33	30.92	92	143	Broker	MD, NM, CMC, CL, OP, EP, EE, CPH, BC, BA, TA
M3	13	26	8	31	5	13.33	10.26	94	70	Broker	ME, AFAHF, AO, CR, BM
M4	20	22	20	30	7	20.00	32.61	146	62	Main Benefit	SS, RE, SL, NBF, FB, HA, CP
Period B: the bull stock market											
M1	4	20	19	38	3	6.67	4.94	46	77	Main Spillover	TP, BC, AO
M2	14	46	52	101	8	23.33	21.60	122	167	Broker	PU, AFAHF, CR, ME, CM, BM, LM, NM
M3	15	35	37	70	7	20.00	23.57	153	120	Main Benefit	MD, CMC, ND, CP, CL, EP, NBF
M4	17	67	82	140	13	40.00	45.75	209	166	Main Benefit	RE, OP, PB, TA, CPH, HA, EE, ET, SL, SS, FB, AM, BA
Period C: the bear stock market											
M1	96	45	15	6	11	33.33	59.26	178	66	Main Benefit	TP, CL, EE, AO, PB, OP, ME, AM, BC, RE, FB
M2	32	11	3	1	5	13.33	23.40	108	36	Main Benefit	CP, ND, SS, BA, HA
M3	41	22	3	1	5	13.33	4.48	42	64	Main Spillover	ET, CR, CMC, MD, CM
M4	105	41	24	21	10	30.00	10.99	8	170	Main Spillover	PU, NM, BM, LM, AFAHF, SL, NBF, TA, CPH, EP
Period D: the China-US trade friction											
M1	66	59	37	33	9	26.67	33.85	182	129	Main Benefit	TP, MD, CP, ET, PB, RE, FB, BC, CMC

M2	62	41	28	23	8	23.33	26.62	158	113	Main Benefit	BM, EP, CR, NM, NBF, CM, CL, AFAHF
M3	71	57	44	46	8	23.33	20.18	88	174	Main Spillover	PU, LM, CPH, TA, ME, BA, SS, AM
M4	49	42	23	23	6	16.67	16.79	102	114	Bilateral Spillover	OP, AO, EE, ND, SL, HA

Period E: the COVID-19 pandemic

M1	1	7	0	1	7	20.00	11.11	18	8	Main Spillover	TP, OP, LM, AFAHF, CPH, RE, ET
M2	2	15	20	10	7	20.00	31.91	70	32	Main Benefit	HA, SS, EE, CMC, BC, NM, PB
M3	10	36	11	7	11	33.33	17.19	37	53	Main Spillover	MD, ND, FB, PU, CL, BA, NBF, TA, AM, SL, AO
M4	6	27	17	12	6	16.67	19.35	18	50	Bilateral Spillover	BM, EP, CR, ME, CM, CP

Note: This table presents the detailed outcomes of a block model analysis on the multidimensional risk contagion networks across five distinct periods: (A) the global financial crisis, (B) the bull stock market, (C) the bear stock market, (D) the China-US trade friction, and (E) the COVID-19 pandemic. For each period, the table shows how industry nodes are partitioned into blocks (e.g., M1, M2) and reports on their interaction characteristics. The columns detail the “Receiving relationship” (links from other blocks), “Number of members”, “Expected” and “Actual internal-relation ratio (%)”, total “Receive links from outside” and “Emit links to outside”, the assigned “Block Roles” (Main Benefit, Main Spillover, Bilateral Spillover, or Broker), and the specific “Industry Nodes” within each block. Industry abbreviations are defined in Appendix A (Table A1).

The number of links received from or emitted to the outside is calculated by summing the corresponding columns or rows in the network’s adjacency matrix, excluding intra-block connections. The “Expected internal-relation ratio” is calculated as (number of members in block - 1) / (total nodes in network - 1). The “Actual internal-relation ratio” is the number of internal links divided by the total links involving the block. The analysis was conducted using the CONCOR algorithm with a maximum cut depth of 2 and a convergence threshold of 0.2. The results reveal a highly dynamic network structure, where the roles and compositions of the blocks shift significantly in response to different economic conditions; for instance, the presence of two distinct “Broker” blocks during the global financial crisis (A) is a unique feature not observed in the other periods, which are primarily characterized by “Main Spillover” and “Main Benefit” roles.

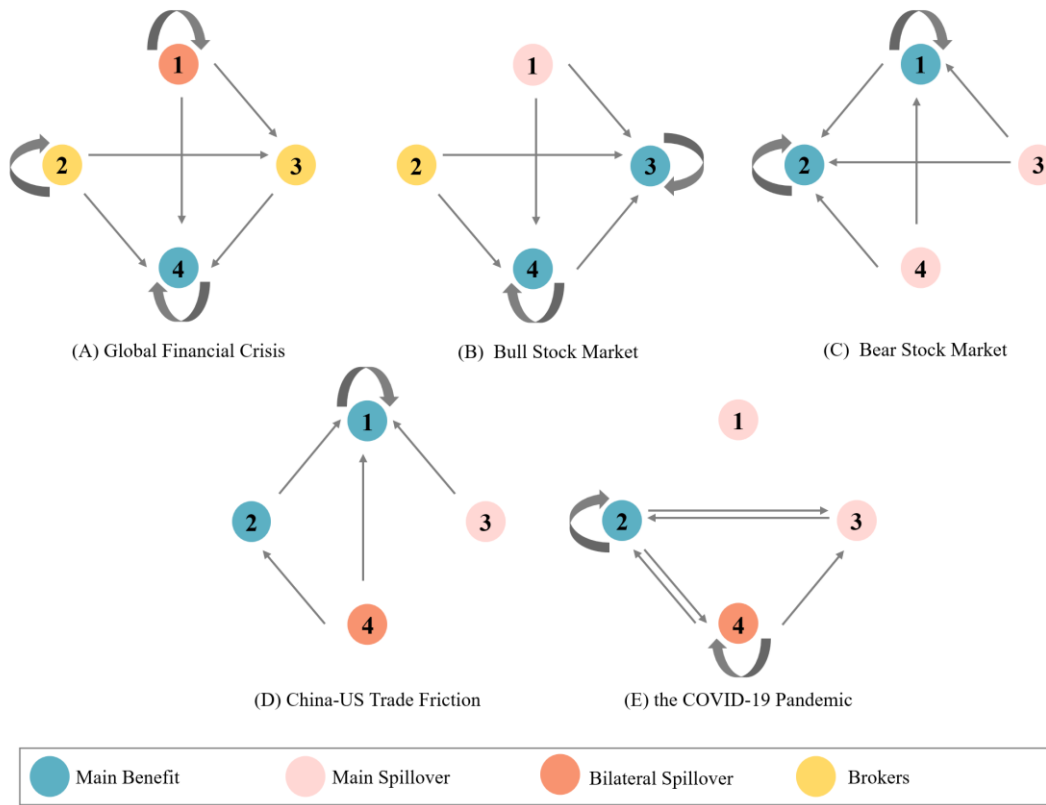
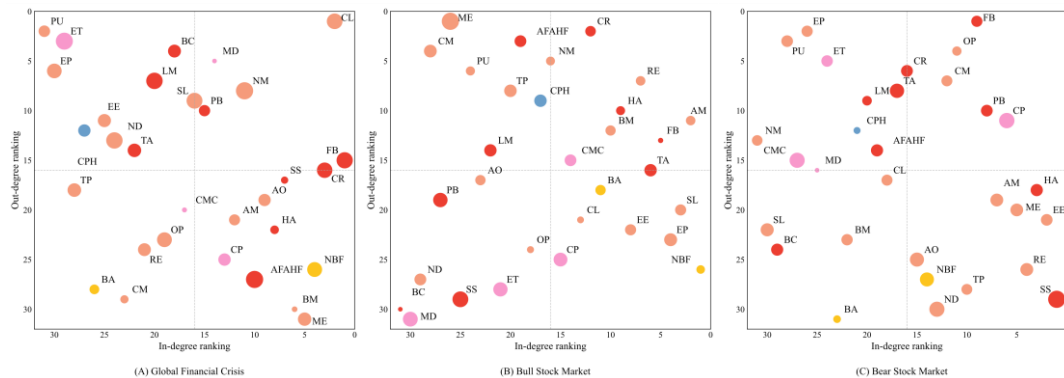


Fig. 5. Risk spillover transmission mechanisms among four blocks.

Note: This figure presents five network diagrams that illustrate the evolution of risk spillover pathways and the functional roles of four industry blocks across different economic periods. In each diagram, arrows denote the direction of risk transmission. The nodes are color-coded to represent their dominant role: Main Benefit (teal), indicating a net receiver of risk; Main Spillover (light pink), indicating a net transmitter of risk; Bilateral Spillover (orange), representing nodes with significant two-way risk exchange; and Brokers (yellow), which facilitate risk transfer between other nodes. The five distinct periods analyzed are (A) the global financial crisis, (B) the bull stock market, (C) the bear stock market, (D) the China-US trade friction, and (E) the COVID-19 pandemic. The analysis reveals that the network's structure is highly sensitive to prevailing economic conditions, shifting from a densely interconnected system with clear broker roles during the systemic crisis of 2008 (A) to sparser and more fragmented configurations during subsequent market phases and shocks (B-E).



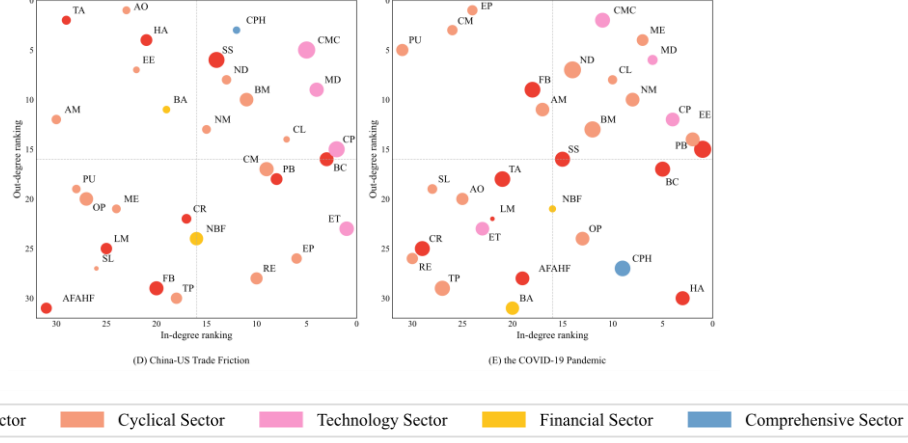


Fig. 6. Node role identification in the multidimensional stock market risk contagion network.

Note: This figure presents five scatter plots that classify the roles of industry nodes based on their in-degree and out-degree rankings within the risk contagion networks across different economic periods. The coordinate system is divided into four quadrants, each representing a distinct functional role: the top-left quadrant identifies ‘Risk Drivers’ (high out-degree, low in-degree); the top-right, ‘Risk Propagators’ (high in-degree and out-degree); the bottom-left, ‘Risk Isolates’ (low in-degree and out-degree); and the bottom-right, ‘Risk Recipients’ (high in-degree, low out-degree). Each circle represents an industry node, with its size reflecting its systemic importance. The nodes are color-coded to denote their sector: Consumption Sector (red), Cyclical Sector (light orange), Technology Sector (pink), Financial Sector (yellow), and Comprehensive Sector (blue). Industry abbreviations are defined in Appendix A (Table A1). The five plots correspond to distinct periods: (A) the global financial crisis, (B) the bull stock market, (C) the bear stock market, (D) the China-US trade friction, and (E) the COVID-19 pandemic. A key observation is the dynamic nature of these roles: for instance, financial sector nodes (yellow) are prominent Risk Recipients and Propagators during the global financial crisis (A) and the bear market (C), while consumption sector nodes (red) consistently act as major Risk Propagators across most periods.

4.3. Analysis of multidimensional risk contagion mechanisms

4.3.1. Model specification and indicator selection

To systematically investigate the multidimensional risk contagion intensity in stock markets under the interplay of cross-industrial mutual effects, intra-industrial self-driven effects, and multi-level economic factors, this study constructs a network vector autoregressive (NAR) model for dynamic risk contagion. The model builds on the coupled multidimensional risk contagion network framework established earlier and draws methodological insights from the network vector autoregressive model proposed by Zhu et al. (2017). The general specification of the NAR model is formulated as:

$$Y_{i,t} = \alpha + \beta \sum_{j=1}^N w_{ij,t-1} Y_{j,t-1} + \gamma Y_{i,t-1} + Z_{i,t-1}^T \delta + \varepsilon_{i,t} = X_{i,t-1}^T \theta + \varepsilon_{i,t} \quad (17)$$

where the explanatory variable $X_{i(t-1)} = (1, \sum_{j=1}^N w_{ij,t-1} Y_{j,t-1}, Y_{i,t-1}, Z_{i,t-1}^T)^T \in R^{p+3}$ includes network spillovers, autoregressive terms, and control variables. The parameter vector $\theta =$

$(\theta_j)^\top = (\alpha, \beta, \gamma, \delta^\top)^\top \in R^{p+3}$ is estimated via ordinary least squares (OLS). Furthermore, let $X_t = (X_{1t}, X_{2t}, \dots, X_{Nt})^\top \in R^{N \times (p+3)}$, then $Y_t = X_{t-1}\theta + \varepsilon_t$. The OLS estimator is then derived as:

$$\hat{\theta} = \left(\sum_{t=1}^T X_{t-1}^\top X_{t-1} \right)^{-1} \sum_{t=1}^T X_{t-1}^\top Y_t \quad (18)$$

The NAR model is operationalized through three specifications to capture distinct dimensions of dynamic risk contagion:

$$OutRisk_{i,t} = \alpha_1 + \beta_1 \sum_{j=1}^N w_{ij,t-1} OutRisk_{j,t-1} + \gamma_1 OutRisk_{i,t-1} + \delta_1 Z_{i,t-1} + \lambda_{industry} + \varepsilon_{i,t} \quad (19)$$

$$InRisk_{i,t} = \alpha_2 + \beta_2 \sum_{j=1}^N w_{ij,t-1} InRisk_{j,t-1} + \gamma_2 InRisk_{i,t-1} + \delta_2 Z_{i,t-1} + \lambda_{industry} + \varepsilon_{i,t} \quad (20)$$

$$Risk_{i,t} = \alpha_3 + \beta_3 \sum_{j=1}^N w_{ij,t-1} Risk_{j,t-1} + \gamma_3 Risk_{i,t-1} + \delta_3 Z_{i,t-1} + \lambda_{industry} + \varepsilon_{i,t} \quad (21)$$

where $i, j = 1, \dots, N$ denote industry indices, t represents the weekly time index, and w_{ij} corresponds to elements of the adjacency matrix derived from the coupled network. The term $\lambda_{industry}$ captures industry-specific fixed effects, while $\varepsilon_{i,t}$ denotes the idiosyncratic error term. To ensure statistical robustness, cluster-robust standard errors are estimated at the industry level to address potential heteroskedasticity and intra-industry correlation in the regression coefficients.

The dependent variables *OutRisk*, *InRisk*, and *Risk* represent the output intensity, input intensity, and overall intensity of multidimensional dynamic risk contagions in stock markets, respectively, which are hypothesized to be influenced by three primary factors. First, the cross-industrial interaction-aggregation effect (β) quantifies the aggregate impact of risk contagion intensities across all N industries on the contagion intensity of a single industry. If $\beta = 0$, the risk contagion intensity of individual industries operates independently or exhibits mutually offsetting effects, indicating no systemic risk interdependence. A negative β ($\beta < 0$) suggests a risk dispersion phenomenon, where contagion intensities across industries mutually attenuate, reducing cross-industrial vulnerability. Conversely, a positive β ($\beta > 0$) signifies risk aggregation, where contagion intensities synergistically amplify within the multidimensional risk network, leading to system-wide escalation. Second, the intra-industrial self-driven effect (γ) measures the autoregressive influence of the one-period lagged risk contagion intensity within industry i on its current intensity. A higher γ reflects stronger temporal persistence in risk propagation. Specifically, $\gamma > 0$ indicates that risk accumulation within an industry exhibits inertia, persisting over time rather than dissipating rapidly, which implies that historical risk exposures propagate nonlinearly into future periods, sustaining prolonged systemic impacts.

Drawing on Prokopczuk et al. (2019) and Yang et al., (2021), this study selects a suite of control variables across three dimensions: macroeconomic conditions, macro-financial environments, and

industry-specific characteristics. Macroeconomic conditions encompass five aspects: RMB exchange rate volatility (*FXVol*), macroeconomic prosperity index (*MacroIndex*), M2 growth rate (*M2Growth*), investor sentiment (*InvSent*), and financialization degree (*FinProp*). Macro-financial environments include four components: term spread (*TermSprd*), credit spread (*CreditSprd*), TED spread (*TEDSprd*), and default spread (*DefSprd*). Industry-specific characteristics cover three dimensions: real estate prosperity (*HouseIndex*), industry trading volume (*Trade*), and industry price-to-earnings ratio (*PERatio*). Detailed definitions and calculation methods for these variables are provided in Table 4, while descriptive statistics are summarized in Table 5. All continuous variables are winsorized at the 1% level to mitigate extreme value influences.

Table 4

Variable definitions and calculation methods.

Variable Type	Variable Name	Definition	Calculation Method
Dependent Variables	<i>OutRisk</i>	Risk contagion output intensity	Average of adjacency matrix elements in the row corresponding to each sector, multiplied by 100
	<i>InRisk</i>	Risk contagion input intensity	Average of adjacency matrix elements in the column corresponding to each sector, multiplied by 100
	<i>Risk</i>	Risk contagion overall intensity	Average of adjacency matrix elements in both rows and columns for each sector, multiplied by 100
Explanatory Variables	<i>Neted_OutRisk</i>	Inter-industrial risk interaction output intensity	Cross-multiplication of the five-dimensional risk adjacency matrix with row-wise averages per sector
	<i>Neted_InRisk</i>	Inter-industrial risk interaction input intensity	Cross-multiplication of the five-dimensional risk adjacency matrix with column-wise averages per sector
	<i>Neted_Risk</i>	Inter-industrial risk interaction overall intensity	Cross-multiplication of the five-dimensional risk adjacency matrix with row and column averages
Control Variables	<i>FXVol</i>	RMB exchange rate volatility	Weekly realized volatility of daily RMB/USD exchange rate changes, scaled by 100
	<i>MacroIndex</i>	Macroeconomic prosperity index	Coincident index from the macroeconomic prosperity index system
	<i>M2Growth</i>	M2 growth rate	growth rate of M2 money supply
	<i>InvSent</i>	Investor sentiment	Investor Sentiment Index (ISI) derived via PCA from closed-end fund discounts, IPO volume, first-day returns, consumer confidence, new account openings, and trading volume, with macroeconomic factors removed
	<i>FinProp</i>	Financialization degree	Percentage of financial sector value-added (constant price) to GDP (constant price)
	<i>TermSprd</i>	Term spread	Difference between 10-year and 1-year government bond yields (daily)
	<i>CreditSprd</i>	Credit spread	Difference between 10-year AAA corporate bond yields and 10-year government bond yields (daily)

	<i>TEDSprd</i>	TED spread	Difference between 1-year SHIBOR and 1-year government bond yields (daily)
	<i>DefSprd</i>	Default spread	Difference between 10-year A-rated and AAA-rated corporate bond yields (daily)
	<i>HouseIndex</i>	Real estate prosperity	National real estate prosperity index
	<i>Trade</i>	Industry trading volume	Natural logarithm of total trading volume in the industry
	<i>PEratio</i>	Industry PE ratio	Market-value-weighted average price-to-earnings ratio (PE) of firms within the industry

Note: This table shows the definitions of the main variables in the regression models.

Table 5

Descriptive statistics of variables.

Variable Type	Variable Name	N	Mean	SD	Min	p50	Max
Dependent Variables	<i>OutRisk</i>	24,986	45.265	10.904	22.347	44.878	70.769
	<i>InRisk</i>	24,986	45.264	11.352	21.183	44.871	72.840
	<i>Risk</i>	24,986	45.268	9.387	24.459	45.190	66.955
Explanatory Variables	<i>Neted_OutRisk</i>	24,986	6.512	2.411	2.191	6.266	12.855
	<i>Neted_InRisk</i>	24,986	6.759	2.456	2.342	6.499	13.234
	<i>Neted_Risk</i>	24,986	6.635	2.433	2.265	6.376	13.057
Control Variables	<i>FXVol</i>	24,986	0.512	0.362	0.007	0.452	1.763
	<i>MacroIndex</i>	24,986	99.125	4.511	88.030	98.800	114.790
	<i>M2Growth</i>	24,986	13.563	4.911	8.000	12.800	29.300
	<i>InvSent</i>	24,986	0.361	1.396	-1.830	0.100	5.380
	<i>FinProp</i>	24,986	6.828	1.193	4.831	7.169	8.250
	<i>TermSprd</i>	24,986	0.831	0.480	0.100	0.704	2.188
	<i>CreditSprd</i>	24,986	1.222	0.332	0.447	1.224	1.930
	<i>TEDSprd</i>	24,986	0.989	0.545	0.172	0.869	2.578
	<i>DefSprd</i>	24,986	4.689	3.797	-5.870	5.920	7.950
	<i>HouseIndex</i>	24,986	99.274	3.555	92.630	100.510	106.000
	<i>Trade</i>	24,986	15.928	1.128	12.876	15.947	18.427
	<i>PEratio</i>	24,986	35.956	22.715	6.150	31.330	126.030

Note: This table shows the descriptive characteristics for the main variables. The sample consists of 24,986 industry-week observations from 2007.01.16 to 2023.12.25. Definitions of the variables are provided in Table 4.

4.3.2. Macroeconomic drivers of the financial-driven channel

Having established that entity-driven and financial-driven channels exhibit distinct activation patterns and transmission mechanisms, we now investigate what macroeconomic and institutional factors render the financial-driven channel particularly fragile and explosive during crises. The NAR model allows us to identify these structural vulnerabilities by controlling for network effects and isolating the impacts of China's macroeconomic environment.

Table 6 presents the NAR empirical results. Beyond the network interaction effects, two macroeconomic variables exhibit particularly robust positive impacts on risk contagion intensity: financialization degree (*FinProp*) and real estate prosperity (*HouseIndex*). The financialization degree consistently shows significant positive coefficients across all specifications—0.980, 0.738, and 0.880 (all significant at the 1% level) for risk output, input, and overall intensities respectively. This indicates that as China's financial sector expands relative to GDP, systemic risk amplification intensifies. The mechanism operates through multiple channels. First, higher financialization concentrates leverage in financial institutions, creating potential for cascading failures when deleveraging occurs. Second, financialization diverts resources from productive sectors toward speculative activities, weakening the real economy's shock-absorbing capacity. Third, complex financial instruments proliferate under high financialization, creating opaque interconnections that facilitate rapid contagion propagation once tail risks materialize.

The positive influence of real estate prosperity (*HouseIndex*) on risk contagion fundamentally stems from the dual financial-real attributes of China's property sector and the structural features of its economy. First, high-leverage-driven risk concentration: During periods of rising real estate prosperity, developers accelerate land acquisitions and project expansions, relying heavily on bank development loans, trust financing, and presale proceeds. Simultaneously, homebuyers increase leverage, leading to a rapid expansion of the financial system's exposure to real estate risks. Second, resource diversion and economic structural fragility: High real estate prosperity attracts excessive capital inflows, crowding out financing opportunities for productive sectors such as manufacturing and technology. This reinforces financialization, diverting financial resources from productive investments to real estate speculation and eroding the endogenous growth momentum of the real economy.

Importantly, financialization and real estate prosperity interact to form a “finance-real estate closed loop.” Risks become interlinked across markets through complex financial instruments such as Commercial Mortgage Backed Securities and supply chain asset-backed securities, resulting in concealed and hard-to-monitor contagion pathways. Notably, periods of high real estate prosperity are not immune to risks but rather stages of latent risk accumulation and accelerated cross-industry transmission. The synergy between real estate prosperity and financialization reflects the deepening of economic financialization—precisely the structural vulnerability that renders the financial-driven channel explosively fragile.

These findings complete the evidentiary chain for our dual-track hypothesis. The dynamic and static analyses established that the financial-driven channel activates explosively during crises while remaining dormant during normal periods, transmitting risks through leverage-based linkages captured by tail risk networks. The NAR results reveal why this channel is so fragile: China's macroeconomic structure—characterized by high financialization and a real estate-centric credit

creation system—provides the fuel for tail risk eruptions. When a spark ignites, such as the 2015 deleveraging policy, the accumulated leverage and interconnected exposures combust into system-wide contagion. In contrast, the entity-driven channel operates continuously through supply chain linkages, reflected in persistent price risk network structures, and proves less sensitive to these macro-financial vulnerabilities. This structural distinction explains the divergent activation patterns, transmission mechanisms, and crisis sensitivities documented across our analyses.

Table 6

NAR results of the influencing factors of risk contagion intensity.

	(1)	(2)	(3)	(4)	(5)	(6)
	Risk contagion output intensity		Risk contagion input intensity		Risk contagion overall intensity	
<i>Neted_OutRisk</i>	0.630*** (0.098)	0.557*** (0.095)				
<i>L_OutRisk</i>	0.440*** (0.022)	0.464*** (0.022)				
<i>Neted_InRisk</i>			0.180*** (0.042)	0.128*** (0.044)		
<i>L_InRisk</i>			0.571*** (0.011)	0.588*** (0.012)		
<i>Neted_Risk</i>					0.045 (0.051)	0.004 (0.052)
<i>L_Risk</i>					0.594*** (0.017)	0.609*** (0.017)
<i>FXVol</i>	1.830*** (0.210)	1.832*** (0.199)	1.820*** (0.278)	1.817*** (0.264)	1.881*** (0.175)	1.877*** (0.169)
<i>MacroIndex</i>	-0.199*** (0.017)	-0.196*** (0.017)	-0.193*** (0.018)	-0.191*** (0.017)	-0.202*** (0.011)	-0.200*** (0.011)
<i>M2Growth</i>	0.166*** (0.038)	0.142*** (0.037)	0.139*** (0.048)	0.133*** (0.048)	0.159*** (0.025)	0.144*** (0.024)
<i>InvSent</i>	-0.194** (0.078)	-0.266*** (0.071)	-0.307*** (0.063)	-0.298*** (0.061)	-0.247*** (0.041)	-0.278*** (0.040)
<i>FinProp</i>	0.980*** (0.204)	0.861*** (0.200)	0.738*** (0.238)	0.750*** (0.227)	0.880*** (0.147)	0.825*** (0.143)
<i>TermSprd</i>	0.595*** (0.215)	0.642*** (0.209)	0.586** (0.226)	0.605** (0.230)	0.580*** (0.142)	0.609*** (0.138)
<i>CreditSprd</i>	1.439*** (0.295)	1.579*** (0.281)	1.656*** (0.327)	1.646*** (0.312)	1.561*** (0.174)	1.621*** (0.163)
<i>TEDSprd</i>	-0.057 (0.189)	-0.112 (0.195)	-0.050 (0.217)	-0.077 (0.220)	-0.025 (0.115)	-0.061 (0.120)
<i>DefSprd</i>	-0.172*** (0.046)	-0.168*** (0.045)	-0.188*** (0.043)	-0.177*** (0.043)	-0.180*** (0.028)	-0.174*** (0.029)

<i>HouseIndex</i>	0.160***	0.158***	0.165***	0.161***	0.174***	0.171***
	(0.032)	(0.032)	(0.035)	(0.035)	(0.022)	(0.022)
<i>Trade</i>	-0.306**	-0.129	0.230	0.121	-0.064	-0.017
	(0.131)	(0.078)	(0.150)	(0.148)	(0.113)	(0.068)
<i>PEratio</i>	-0.005	0.004	0.002	0.006	-0.001	0.005
	(0.005)	(0.006)	(0.006)	(0.005)	(0.004)	(0.003)
Constant	18.945***	15.986***	7.850**	9.229**	11.410***	10.531***
	(4.097)	(3.835)	(3.523)	(3.706)	(2.522)	(2.062)
Industry FE	YES	NO	YES	NO	YES	NO
Observations	24986	24986	24986	24986	24986	24986
Adjusted R ²	0.389	0.386	0.411	0.407	0.438	0.436

Note: Table 6 reports the results of tests examining the cross-industrial interaction-aggregation effect and the intra-industrial self-driven effect. The sample consists of 24,986 industry-week observations from 2007.01.16 to 2023.12.25. All variables are defined in Table 4. Standard errors are clustered by industry, shown in parentheses. All continuous variables are winsorized at 1% and 99%. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

4.3.3. Robustness tests

In our baseline regression, the multidimensional risk coupled networks are constructed at a weekly frequency using a 5-trading-day step length. Recognizing that networks built upon different step lengths are designed to capture inter-industry risk contagion over varying time horizons, we conduct a key robustness check to validate our baseline results and ensure they are not contingent on this specific methodological choice. Specifically, we reconstruct the multidimensional coupled networks using alternative step lengths of 10, 20, 30, 50, 100 and 120 trading days. All frequency-dependent variables are adjusted accordingly. For instance, the RMB exchange rate volatility (*FXVol*) is recalculated as the realized volatility derived from daily RMB-USD exchange rate returns over forward-rolling windows of 10, 20, 30, 50, 100 and 120 trading days, respectively, and then scaled by 100. The industry price-to-earnings ratio (*PEratio*) and industry trading volume (*Trade*), based on the value-weighted price-to-earnings ratio of constituent firms and the natural logarithm of total trading volume, respectively, are recomputed as averages over these corresponding periods. The remaining control variables are invariant to the step length, as they are measured at daily or month-end frequencies, thus requiring no adjustment.

Table 7 represents the regression results using different step lengths for network construction. First, we use shorter-term step lengths of 10 and 20 trading days. The results, reported in Table 7 Panel A, show that the coefficients of main independent variables remain similar to our baseline regressions, confirming our main findings. Second, we employ medium-term step lengths of 30 and 50 trading days. We repeat the baseline regressions and present the results in Table 7 Panel B. The estimations yield similar results, with the key coefficients retaining their sign and statistical significance. Third, we use much longer step lengths of 100 and 120 trading days to construct the networks, the results of which are reported in Table 7 Panel C. We find that even at these lower frequencies, the baseline relationships hold: the coefficients on the input and output intensity network effects are still positive and statistically significant, while the non-significance of the

overall intensity network effect remains. Overall, these tests confirm that our main conclusions are robust to a wide range of network construction frequencies.

Table 7

Robustness tests: varying the step length for network construction.

Panel A: Regression results using alternative step lengths of 10 or 20 trading days						
	(1)	(2)	(3)	(4)	(5)	(6)
	step = 10			step = 20		
Dependent var.	<i>OutRisk</i>	<i>InRisk</i>	<i>Risk</i>	<i>OutRisk</i>	<i>InRisk</i>	<i>Risk</i>
<i>Neted_OutRisk</i>	0.535***			0.650***		
	(0.122)			(0.163)		
<i>L_OutRisk</i>	0.457***			0.429***		
	(0.028)			(0.036)		
<i>Neted_InRisk</i>		0.162***			0.183***	
		(0.046)			(0.056)	
<i>L_InRisk</i>		0.567***			0.565***	
		(0.013)			(0.014)	
<i>Neted_Risk</i>			0.044			0.061
			(0.062)			(0.079)
<i>L_Risk</i>			0.588***			0.583***
			(0.020)			(0.024)
Constant	35.078***	22.234***	26.694***	25.265***	8.960*	14.741***
	(5.091)	(3.844)	(2.898)	(6.367)	(4.987)	(4.084)
Controls	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES
Observations	12462	12462	12462	6231	6231	6231
Adjusted R ²	0.407	0.429	0.466	0.402	0.429	0.467
Panel B: Regression results using alternative step lengths of 30 or 50 trading days						
	(1)	(2)	(3)	(4)	(5)	(6)
	step = 30			step = 50		
Dependent var.	<i>OutRisk</i>	<i>InRisk</i>	<i>Risk</i>	<i>OutRisk</i>	<i>InRisk</i>	<i>Risk</i>
<i>Neted_OutRisk</i>	0.762***			1.196***		
	(0.161)			(0.243)		
<i>L_OutRisk</i>	0.416***			0.387***		
	(0.034)			(0.053)		
<i>Neted_InRisk</i>		0.157**			0.405***	
		(0.067)			(0.086)	
<i>L_InRisk</i>		0.582***			0.589***	
		(0.016)			(0.029)	
<i>Neted_Risk</i>			0.139			0.251
			(0.089)			(0.151)
<i>L_Risk</i>			0.575***			0.617***

			(0.025)			(0.046)
Constant	40.649***	25.424***	31.615***	18.281**	0.929	4.207
	(6.631)	(5.975)	(4.227)	(8.484)	(6.479)	(5.117)
Controls	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES
Observations	4154	4154	4154	2480	2480	2480
Adjusted R ²	0.446	0.467	0.520	0.396	0.408	0.433

Panel C: Regression results using alternative step lengths of 100 or 120 trading days

	(1)	(2)	(3)	(4)	(5)	(6)
	step = 100			step = 120		
Dependent var.	<i>OutRisk</i>	<i>InRisk</i>	<i>Risk</i>	<i>OutRisk</i>	<i>InRisk</i>	<i>Risk</i>
<i>Neted_OutRisk</i>	1.525***			1.235***		
	(0.365)			(0.359)		
<i>L_OutRisk</i>	0.350***			0.254***		
	(0.078)			(0.077)		
<i>Neted_InRisk</i>		0.452***			0.218*	
		(0.100)			(0.122)	
<i>L_InRisk</i>		0.607***			0.546***	
		(0.032)			(0.032)	
<i>Neted_Risk</i>			0.268			0.105
			(0.180)			(0.173)
<i>L_Risk</i>			0.638***			0.545***
			(0.054)			(0.049)
Constant	-12.547	-28.765***	-26.388***	41.354**	18.228	26.566***
	(11.217)	(9.387)	(6.026)	(17.053)	(12.917)	(9.164)
Controls	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES
Observations	1240	1240	1240	1023	1023	1023
Adjusted R ²	0.435	0.437	0.481	0.463	0.479	0.567

Note: This table reports the results of robustness tests where the step length for constructing the risk networks is varied. Panel A presents regression results using step lengths of 10 and 20 trading days. Panel B uses step lengths of 30 and 50 trading days, and Panel C uses step lengths of 100 and 120 trading days. The dependent variables are *OutRisk* (risk contagion output intensity), *InRisk* (risk contagion input intensity), and *Risk* (risk contagion overall intensity). “L_” denotes the one-period lagged variable. All regressions include the same set of control variables as in the baseline model, defined in Table 4, and control for industry fixed effects. Standard errors are clustered by industry, shown in parentheses. All continuous variables are winsorized at 1% and 99%. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

5. Conclusions

This study examines inter-industry risk contagion in China through a multi-dimensional coupled network framework spanning 2007-2023. By decomposing aggregate risk into five

dimensions and analyzing their distinct transmission patterns, we uncover a dual-track contagion mechanism that operates through fundamentally different channels with heterogeneous temporal dynamics, network topologies, and macroeconomic sensitivities.

Our central finding is that inter-industry risk transmission follows two coexisting yet distinct tracks. The entity-driven channel, proxied by price risk, functions as a structural backbone. It exhibits persistent high edge uniqueness and remarkably stable coupling weights (approximately 0.20) throughout the sample period, reflecting continuous cost and demand propagation along supply chain networks. This structural channel maintains stable sectoral contributions, with cyclical industries consistently dominating transmission regardless of market conditions.

In contrast, the financial-driven channel, proxied by tail risk, remains dormant during tranquil periods but activates explosively during systemic crises. Dynamic coupling weight analysis reveals a critical mechanism for distinguishing crisis types: during the 2015 stock market crash, tail risk weights surged to dominate aggregate connectivity, coinciding with peak edge uniqueness to signal complete channel activation driven by leverage unwinding. Conversely, during the COVID-19 pandemic, a divergence emerged: while tail risk uniqueness rose, coupling weights remained moderate, indicating that the shock propagated primarily through supply chains rather than systemic financial dominance. This evidence confirms that the dominant transmission mechanism switches between entity-based and leverage-amplified pathways depending on whether the shock originates from the real economy or the financial system.

Static case studies further validate this mechanism. The China-US trade friction period exemplifies entity-driven contagion: technology sectors cohesively absorbed risks through supply chain disruptions, transmission followed production linkages, and tail risk networks remained subdued. Conversely, the 2015 crash demonstrates financial-driven contagion: non-bank financial institutions became dominant risk propagators, technology sectors fragmented as leverage exposure superseded production relationships, and bidirectional amplification loops replaced unidirectional supply chain flows. Network autoregressive analysis identifies the macroeconomic roots of this asymmetry. Financialization degree and real estate prosperity significantly amplify tail risk contagion while exerting negligible influence on price risk transmission, indicating that the financial channel's fragility stems from China's structural dependence on leverage and property-backed credit creation.

These findings carry implications for financial regulation and systemic risk management. Our evidence suggests that effective risk prevention requires a dual-track regulatory approach aligned with the distinct characteristics of each contagion channel.

For the entity-driven channel, regulators should establish supply-chain-based early-warning systems leveraging industry price indices and production network centrality metrics. Given this channel's persistent, structural nature, monitoring should focus on routine risk management rather than crisis-specific interventions. Price risk networks maintain high connectivity across all periods, with cyclical industries consistently serving as transmission hubs. Accordingly, surveillance systems should track cost propagation along upstream-downstream linkages, identify industries whose production centrality renders them systemically important for real-economy stability, and implement targeted support during supply disruptions to prevent cascading failures through

manufacturing networks. This constitutes continuous, non-crisis-specific oversight aimed at maintaining operational resilience in the real economy.

For the financial-driven channel, regulators must prioritize macroprudential policies targeting financial leverage and real estate financing to prevent tail risk activation. Our NAR results demonstrate that financialization degree and property prosperity significantly amplify contagion intensity, providing fuel that ignites system-wide crises once triggered by deleveraging shocks or panic episodes. Regulatory focus should concentrate on financial institutions' leverage ratios, shadow banking exposures, and real estate credit concentrations. Given this channel's crisis-activated nature, interventions should emphasize preemptive containment of leverage buildup and rapid stabilization when tail dependencies threaten to cascade. During normal periods, macroprudential tools should constrain excessive credit growth and speculative financing. During stress episodes, liquidity provision and orderly deleveraging protocols become critical to preventing the explosive amplification documented in our 2015 crash analysis. This constitutes crisis prevention at its core.

Single-track regulatory frameworks that treat all risk dimensions equivalently will prove inadequate for managing systemic risk in emerging markets characterized by both persistent supply chain interdependencies and episodic financial instabilities. The dual-track approach acknowledges that different shock origins require differentiated policy responses: structural vulnerabilities demand continuous monitoring and gradual adjustment, whereas leverage-driven fragilities necessitate aggressive preemptive action to forestall nonlinear escalation. Our findings from China's experience offer broader lessons for emerging economies undergoing financial deepening while maintaining extensive real-economy production networks, where the coexistence of entity-driven and financial-driven contagion channels creates distinct regulatory challenges that conventional single-dimension frameworks cannot adequately address.

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Appendix A. List of full names and abbreviations of industries

Table A1

List of full names and abbreviations of industries

Full name	Abbreviation	Full name	Abbreviation
Media	MD	Transportation	TP
Electrical Equipment	EE	Coal	CL
Electronics	ET	Beauty and Personal Care	BC
Real Estate	RE	Agriculture, Forestry, Animal Husbandry, and Fishery	AFAHF
Textile and Apparel	TA	Automobile	AM
Non-Bank Finance	NBF	Light Manufacturing	LM
Steel	SL	Commercial and Retail	CR
Public Utilities	PU	Social Services	SS
National Defense	ND	Oil and Petrochemical	OP
Environmental Protection	EP	Food and Beverage	FB
Machinery Equipment	ME	Communications	CMC
Chemical	CM	Pharmaceuticals and Biotechnology	PB
Computer	CP	Banking	BA
Household Appliances	HA	Nonferrous Metals	NM
Building Materials	BM	Comprehensive	CPH
Architectural Ornament	AO		

Note: This table provides the full names and standard abbreviations for all 31 Shenwan Level-1 industries used throughout the analysis. The abbreviations are consistently applied in all tables, figures, and empirical discussions to ensure clarity and conciseness in reporting sector-specific results.

Appendix B. CoVaR estimation procedure using DCC-GARCH models

The VaR for market i at quantile q is defined through the conditional probability:

$$Pr(X^i \leq VaR_q^i) = q \quad (1)$$

where X^i represents the return variable of the industrial stock market i , and VaR_q^i denotes the maximum potential loss at the $1-q$ confidence level. While VaR measures individual market risk exposure, it fails to capture the risk spillover intensity between markets. This limitation motivates the development of CoVaR, which is formally defined through the conditional quantile:

$$Pr(X^j \leq CoVaR_q^{j|i} | X^i = VaR_q^i) = q \quad (2)$$

This formulation quantifies the maximum potential loss in market j at the $1-q$ confidence level, conditional on market i experiencing an extreme risk event ($X^i = VaR_q^i$).

This study employs the DCC-GARCH model to analyze the return series of various stock market industries and the overall market, thereby characterizing the dynamic correlations among industries, based on which the CoVaR measures are further calculated. Engle (2002) proposed the DCC-GARCH model (Dynamic Conditional Correlation Generalized Autoregressive Conditional Heteroskedasticity model), which is particularly useful for examining time-varying correlations between assets. The bivariate DCC-GARCH (1,1) model specification is as follows:

$$r_t = \mu_t + \varepsilon_t \quad (3)$$

$$\mu_t = E(r_t | \varphi_t) \quad (4)$$

where r_t denotes the return series of industries, μ_t represents the intercept term, ε_t is the residual term, and φ_t is the information set available at time t .

$$H_t = D_t R_t D_t \quad (5)$$

$$D_t = \text{diag}\{\sqrt{h_{i,t}}\} \quad (6)$$

$$R_t = \text{diag} Q_t^{-1/2} Q_t \text{diag} Q_t^{-1/2} \quad (7)$$

$$Q_t = (1 - \alpha - \beta) \bar{Q} + \alpha \varepsilon_{t-1} \varepsilon'_{t-1} + \beta Q_{t-1} \quad (8)$$

Here, H_t is the time-varying conditional covariance matrix, D_t is the diagonal matrix of time-varying standard deviations obtained from univariate GARCH models, R_t is the dynamic conditional correlation matrix, Q_t is the conditional covariance matrix, and $\text{diag} Q_t^{-1/2}$ represents the conditional standard deviation matrix of standardized residuals. $\bar{Q} = E[\varepsilon_{t-1} \varepsilon'_{t-1}]$ denotes the estimated unconditional covariance matrix of standardized residuals, while $\varepsilon_t = D_t^{-1} \varepsilon_t$ are the standardized residuals. α and β are the two coefficients of the DCC-GARCH model. α measures the impact of past squared residuals on current conditional correlations, reflecting the “short-term volatility” effect in the market, which represents the immediate market response to shocks. β measures the influence of past conditional correlations on current conditional correlations, reflecting the “long-term memory” characteristic of the market, describing the persistence or stationarity of conditional correlations. The constraint $\alpha + \beta \leq 1$ holds, and values closer to 1 indicate stronger dependence of conditional correlations on historical shocks (short-term) and persistence (long-term).

Defining the return series of the overall stock market as i and that of a specific industry as j , the conditional Value-at-Risk of series j when series i experiences a risk loss of $VaR_{q,t}^i = \Phi^{-1}(q\%)$ can be expressed as:

$$CoVaR_{q,t}^{j|i} = \Phi^{-1}(q\%) \sigma_t^j \sqrt{1 - (\rho_{t,t}^{ij})^2} + \Phi^{-1}(q\%) \rho_{t,t}^{ij} \sigma_t^i \quad (9)$$

where σ_t^i , σ_t^j , and ρ_t^{ij} are the conditional standard deviations of series i and j , and their dynamic correlation coefficient, respectively, all estimated by the DCC-GARCH model.

Appendix C. Multidimensional network coupling method

Consider the following equation:

$$A(R_t - E[R_t]) = \beta F_t + \eta_t \quad (10)$$

where: R_t represents the returns of each node (industrial stock markets); F_t denotes the common factor influencing all nodes' returns, here taken as the first principal component of the nodes' returns; A is a matrix capturing contemporaneous relationships among nodes' returns; η_t is the residual vector.

Assuming A is invertible, the reduced-form equivalent equation is:

$$\begin{aligned} R_t &= E[R_t] + A^{-1}\beta F_t + A^{-1}\eta_t \\ &= E[R_t] + \beta^* F_t + \eta_t^* \end{aligned} \quad (11)$$

Here, the reduced-form residuals $\eta_t^* = A^{-1}\eta_t$ and coefficients $\beta^* = A^{-1}\beta$ are both nonlinear functions of the contemporaneous relationship matrix A^{-1} . Drawing on spatial econometric techniques (Billio et al., 2023), A is expressed as an affine function of the network:

$$A = I - \sum_{d=1}^D \delta_d W_d \quad (12)$$

where the coefficients δ_d , which quantify the degree to which each subnetwork W_d affects returns, serve as subnetwork weights subject to $\sum_{d=1}^D \delta_d = 1$.

To account for the heterogeneous impact of the network on each node, that is, the heterogeneous effects of the network on each node's returns, a diagonal matrix $R = \text{diag}(\rho_1, \rho_2, \dots, \rho_n)$ is introduced:

$$A = I - R \sum_{d=1}^D \delta_d W_d \quad (13)$$

Thus, a multidimensional coupled network model with n nodes and D subnetworks is formalized as:

$$\left[I - R \left(\sum_{d=1}^D \delta_d W_d \right) \right] R_t = E[R_t] + \beta F_t + \eta_t \quad (14)$$

This study adopts an iterative maximum likelihood estimation approach, which represents the established methodology in spatial econometrics, to estimate both the subnetwork weights δ_d and the elements ρ_n of the diagonal matrix. The multidimensional coupled network adjacency matrix is:

$$W^* = \sum_{d=1}^D \delta_d W_d \quad (15)$$

Appendix D. ADF test results for static analysis

Table A2

ADF test results for the logarithmic return series under static analysis

Periods	the global financial crisis	the bull stock market	the bear stock market	the China-US trade friction	the COVID-19 pandemic
Log Return Series	ADF statistic				
Overall Market	-13.677***	-9.992***	-8.358***	-13.671***	-9.064***
TP	-13.619***	-9.149***	-8.352***	-13.517***	-9.095***
MD	-14.098***	-10.598***	-8.339***	-13.310***	-10.111***
PU	-13.946***	-11.050***	-8.134***	-12.911***	-8.972***
AFAHF	-13.871***	-11.545***	-8.963***	-13.011***	-8.789***
PB	-14.940***	-11.633***	-8.517***	-13.906***	-10.166***
CR	-14.370***	-10.660***	-8.088***	-12.938***	-9.238***
ND	-13.780***	-10.134***	-8.436***	-13.089***	-10.341***
CM	-13.997***	-10.555***	-8.465***	-12.944***	-9.199***
HA	-14.270***	-10.960***	-9.032***	-13.795***	-9.362***
BM	-13.947***	-9.691***	-8.686***	-13.096***	-9.153***
AO	-14.428***	-10.000***	-8.210***	-12.786***	-9.562***
RE	-13.527***	-10.611***	-8.050***	-12.854***	-8.959***
NM	-13.150***	-9.784***	-8.086***	-12.570***	-9.121***
ME	-14.027***	-9.923***	-8.702***	-12.947***	-9.444***
AM	-14.194***	-9.825***	-8.920***	-13.795***	-9.236***
CL	-13.762***	-9.574***	-8.367***	-12.906***	-9.142***
EP	-13.755***	-10.586***	-8.571***	-12.245***	-9.646***
EE	-14.484***	-10.199***	-8.230***	-12.762***	-9.790***
ET	-14.148***	-11.319***	-8.450***	-13.182***	-10.006***
OP	-13.506***	-10.301***	-8.495***	-13.523***	-8.661***
SS	-13.890***	-10.699***	-9.188***	-13.079***	-9.755***
TA	-13.759***	-10.137***	-8.467***	-12.961***	-8.745***
CPH	-13.809***	-10.561***	-8.058***	-12.812***	-9.620***
CP	-14.099***	-10.971***	-8.216***	-13.367***	-10.356***
LM	-13.802***	-10.668***	-8.354***	-12.832***	-9.394***
CMC	-13.904***	-10.078***	-8.235***	-12.459***	-10.247***
SL	-14.461***	-9.629***	-8.613***	-12.432***	-9.383***
NBF	-14.107***	-11.134***	-8.695***	-13.917***	-9.612***
FB	-14.361***	-10.342***	-8.956***	-14.070***	-9.097***
BC	-14.177***	-10.630***	-8.289***	-13.668***	-10.790***
BA	-13.878***	-10.046***	-8.378***	-13.417***	-9.388***

Note: *** indicates significance at the 1% level ($p \leq 0.01$), ** indicates significance at the 5% level ($p \leq 0.05$), and * indicates significance at the 10% level ($p \leq 0.10$).